The Macroscopic Shape of Gelfand-Tsetlin Patterns & Free Probability

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Discrete Random Structures

Będlewo

2026

Joint work with

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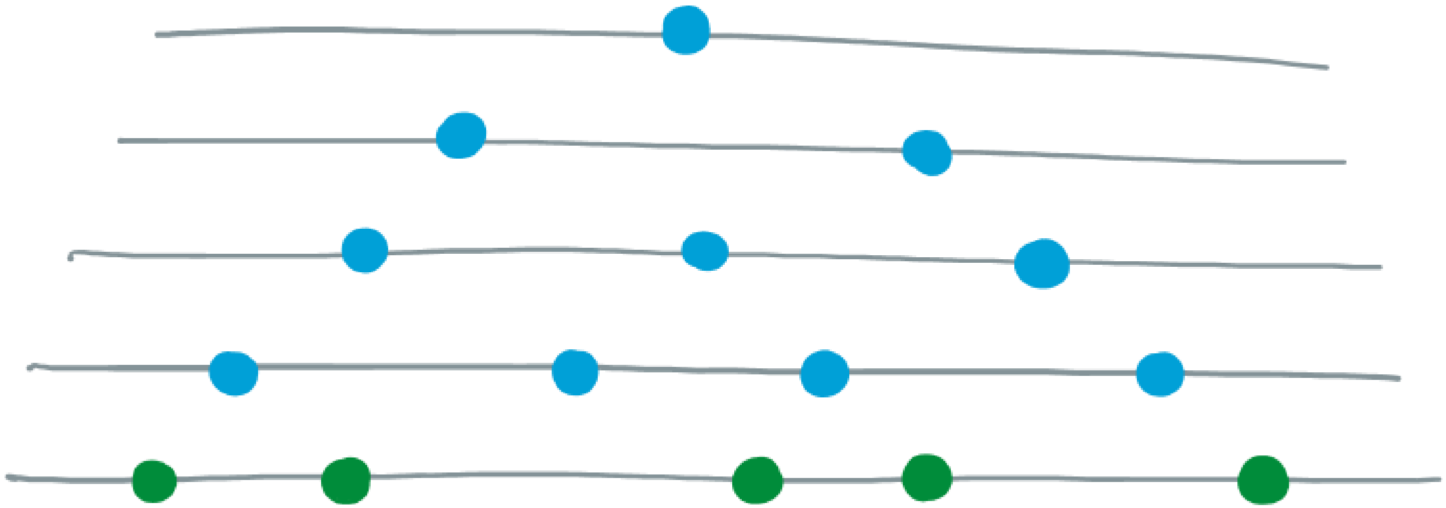
1. Gelfand-Tsetlin Patterns

A Gelfand-Tsetlin Pattern is an array of real numbers

$$(t_{kj})_{1 \le j \le k \le n}$$

satisfying certain inequalities...

A GELFAND-TSETLIN PATTERN IS A COLLECTION OF REAL NUMBERS THAT LOOKS LIKE THIS :



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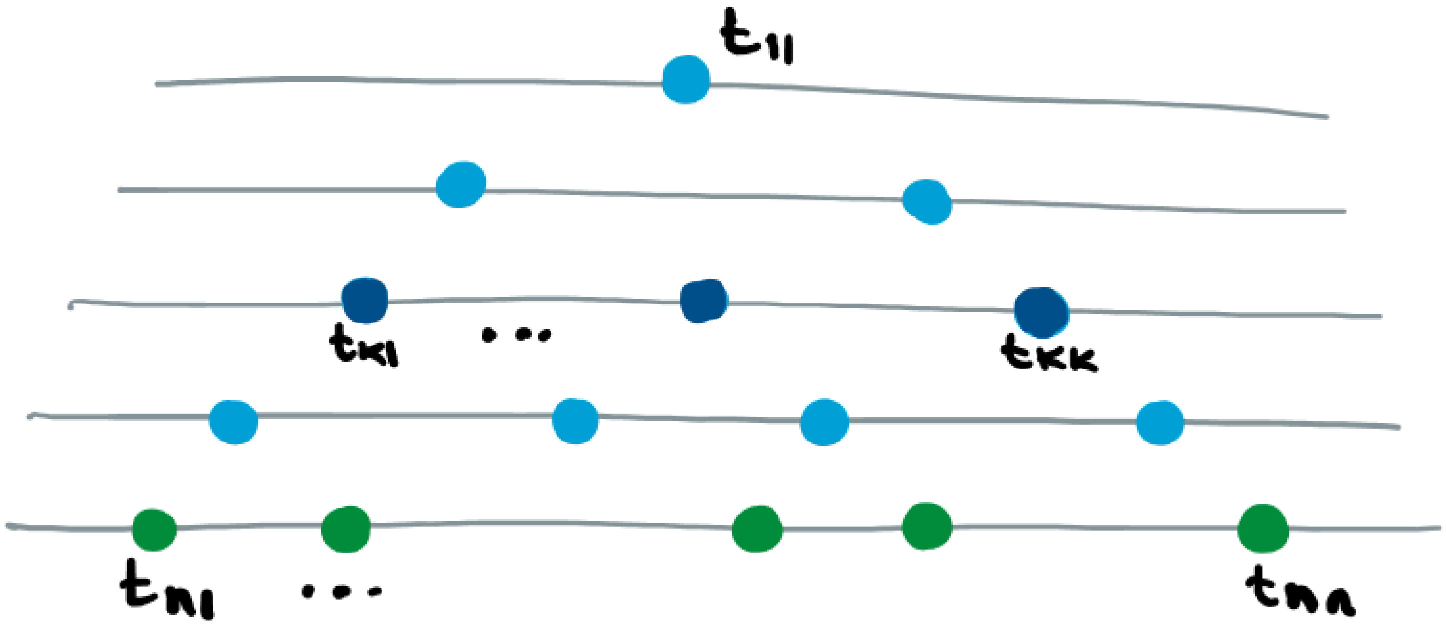

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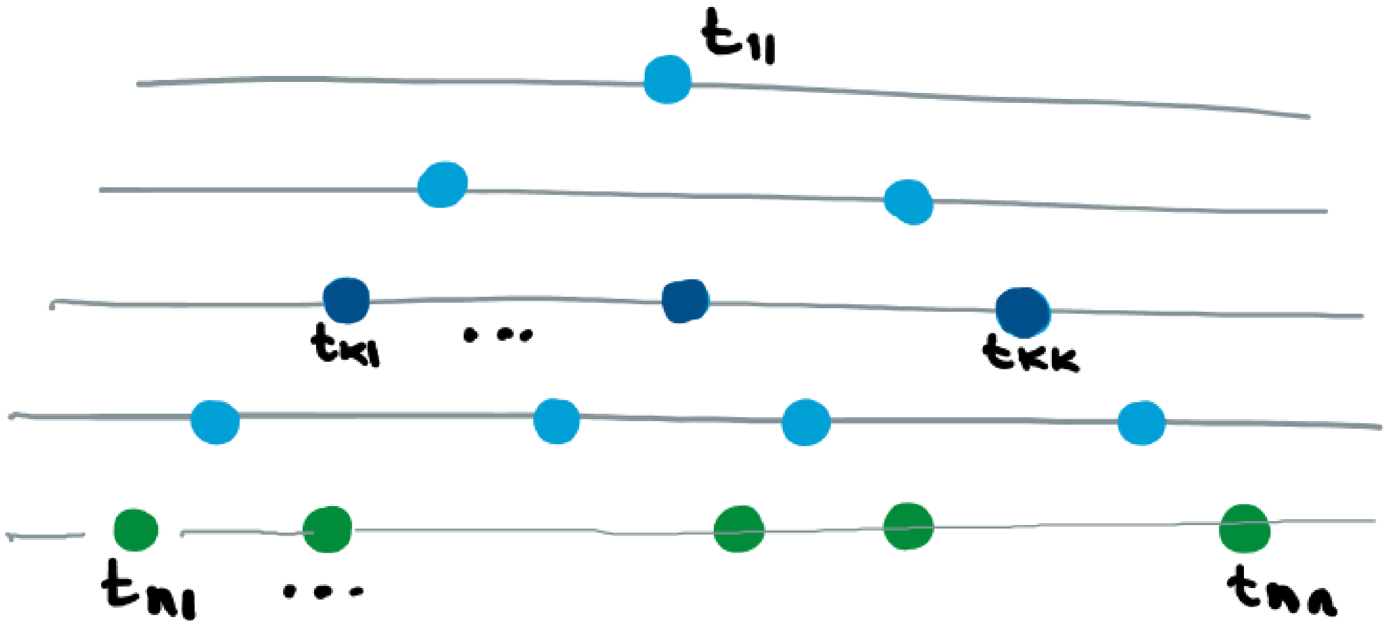
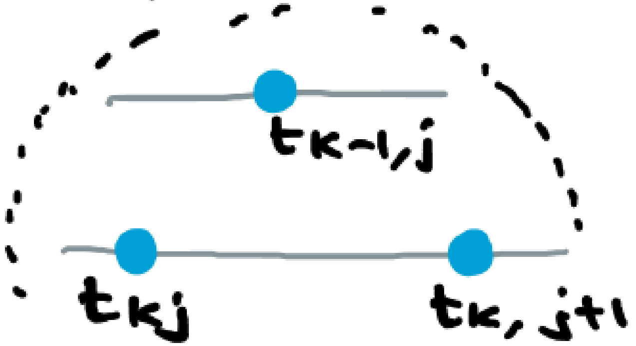
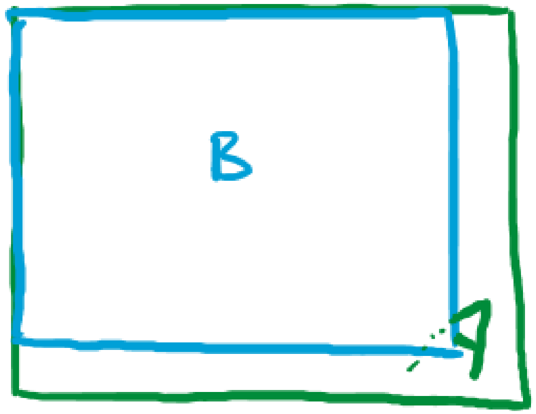


A GT PATTERN IS AN ARRAY $(t_{kj})_{1 \le j \le k \le n}$ OF $t_{kj} \in \mathbb{R}$ SATISFYING INEQUALITIES

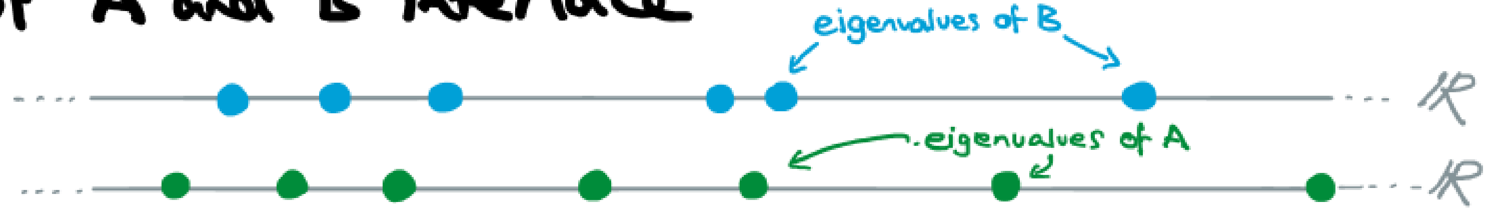


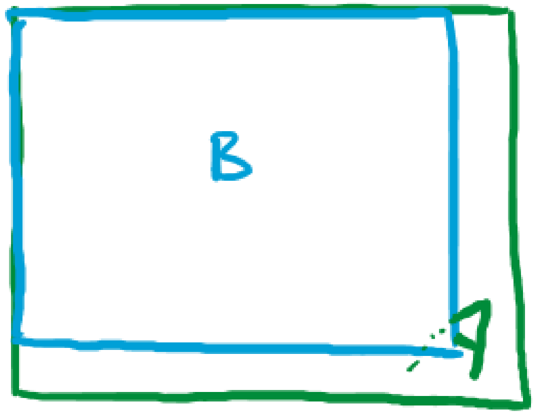
$$t_{kj} \le t_{k-1,j} \le t_{k,j+1}$$



Cauchy's interlacing theorem:

if A is an n-by-n Hermitian matrix, and B is its $(n-1)$-by-$(n-1)$ principal minor, the eigenvalues of A and B interlace

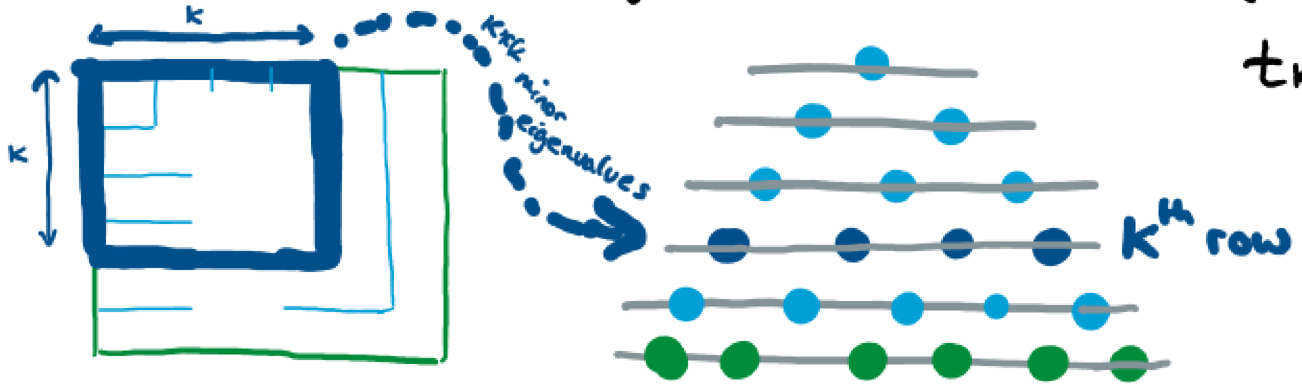




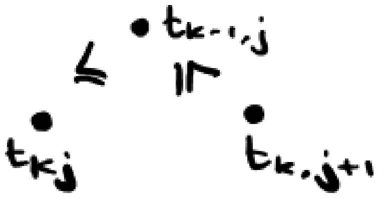
Cauchy's interlacing theorem:

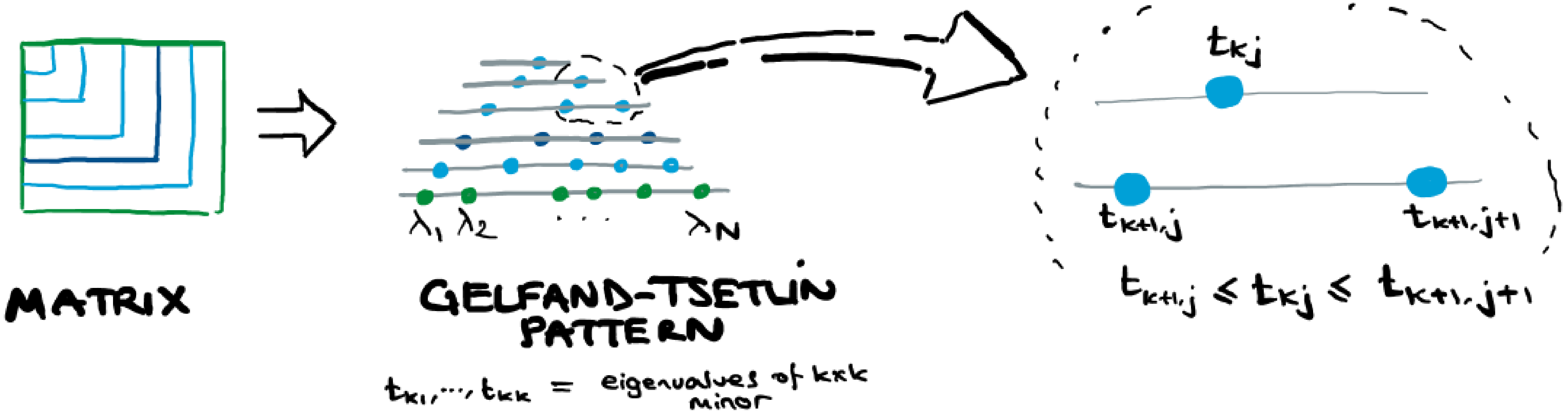
if A is an n-by-n Hermitian matrix, and B is its n-by-n principal minor, the eigenvalues of A and B interlace

We can look at all eigenvalues of successive minors and obtain a **Gelfand-Tsetlin pattern**. $(t_{kj})_{1 \le j \le k \le n}$

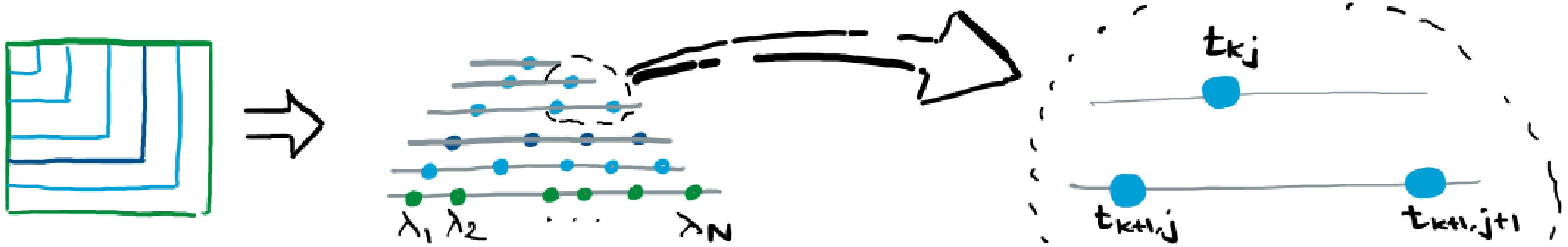


$t_{kj} = j^{th}$ eigenvalue of $k \times k$ minor.





- An $N\times N$ matrix with eigenvalues $\lambda_1 \le \dots \le \lambda_N$ gives rise to a GT-pattern $(t_{kj})_{1\le j\le k\le N}$ with bottom row $t_{Nj}=\lambda_j$.



MATRIX

GELFAND-TSETLIN PATTERN

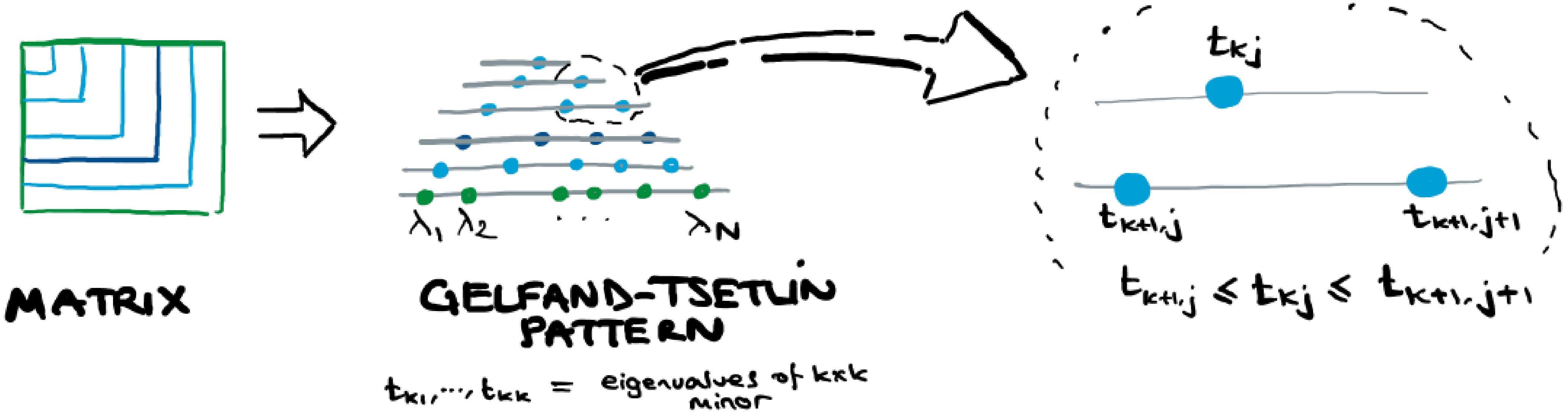
$t_{k1}, \dots, t_{kk}$ = eigenvalues of $k \times k$ minor

$t_{k+1,j} \leq t_{kj} \leq t_{k+1,j+1}$

- AN $N \times N$ MATRIX WITH EIGENVALUES $\lambda_1 \leq \cdots \leq \lambda_N$ GIVES RISE TO A GT-PATTERN $(t_{kj})_{1 \leq j \leq k \leq N}$ WITH BOTTOM ROW $t_{Nj} = \lambda_j$.
- THE SET

$$GT(\lambda_1, \dots, \lambda_N) = \left\{ (t_{kj})_{1 \leq j \leq k \leq N} \text{ GT PATTERN} : \begin{array}{c} t_{Nj} = \lambda_j \\ j = 1 \dots N \end{array} \right\}$$

is A COMPACT SUBSET OF $\mathbb{R}^{N(N+1)/2}$.



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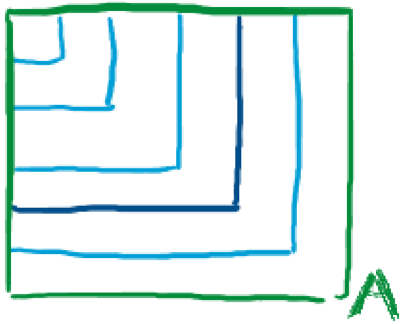
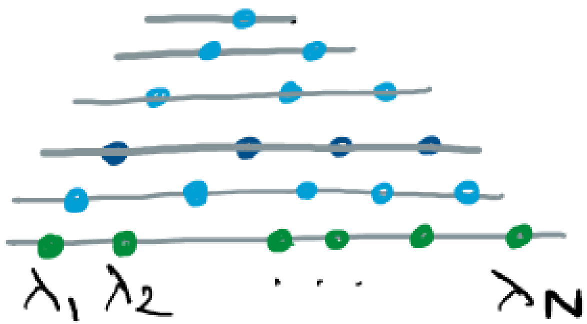
$$GT(\lambda_1, \dots, \lambda_N) = \left\{ (t_{kj})_{1 \leq j \leq k \leq N} \text{ GT PATTERN} : \underset{j=1\dots N}{t_{Nj} = \lambda_j} \right\}$$
is A COMPACT SUBSET OF $\mathbb{R}^{N(N+1)/2}$.
- THUS WE CAN CHOOSE UNIFORMLY FROM $GT(\lambda_1 \cdots \lambda_N)$.



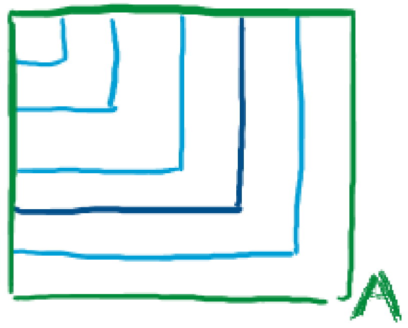
$\Rightarrow$

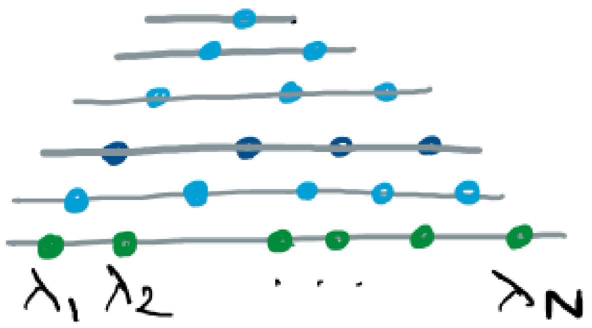


$t_{k1}, \dots, t_{kk} =$ EIGENVALUES OF (RANDOM) $k \times k$ MINOR.

RANDOM MATRIX WITH EIGENVALUES $\lambda_1 \leq \cdots \leq \lambda_N$

RANDOM GT PATTERN WITH BOTTOM ROW $\lambda_1 \leq \cdots \leq \lambda_N$





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<u>RANDOM</u> MATRIX WITH EIGENVALUES $\lambda_1 \leq \cdots \leq \lambda_N$

<u>RANDOM</u> GT PATTERN WITH BOTTOM ROW $\lambda_1 \leq \cdots \leq \lambda_N$

<u>THEOREM</u> (BARYSHNIKOV 1999 / 20TH CENTURY FOLKLORE)

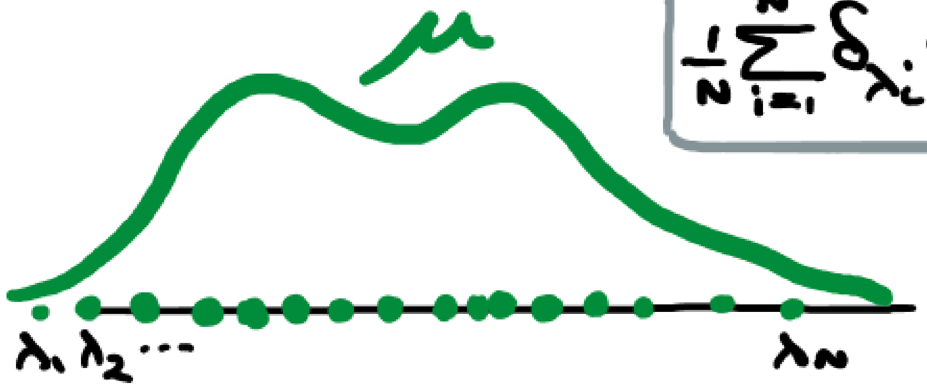
IF A IS UNITARILY INVARIANT RANDOM MATRIX WITH EIGENVALUES

$$\lambda_1 \leq \cdots \leq \lambda_N$$

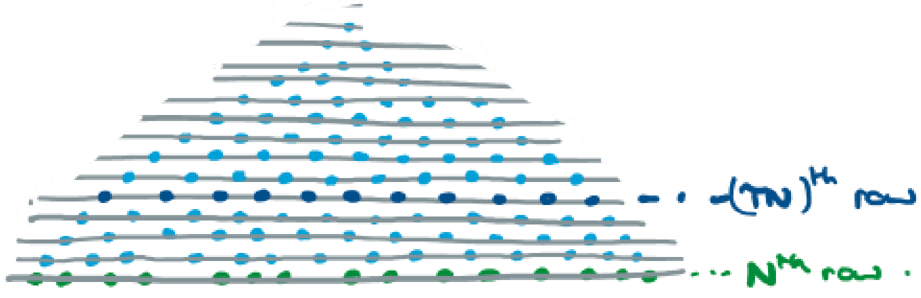
THEN ITS GELFAND-TSETLIN PATTERN IS UNIFORMLY DISTRIBUTED ON SET OF GT PATTERNS WITH BOTTOM ROW $\lambda_1 \leq \cdots \leq \lambda_N$.

FREE PROBABILITY is the study of how the eigenvalues of large random matrices behave under matrix operations (adding, multiplying, taking minors).

TAKE A GIANT RANDOM GELFAND-TSETLIN PATTERN WITH DETERMINISTIC BOTTOM ROW APPROXIMATING PROB LAW μ



$$\boxed{\frac{1}{N}\sum_{i=1}^{N} \delta_{\lambda_i} \approx \mu}$$



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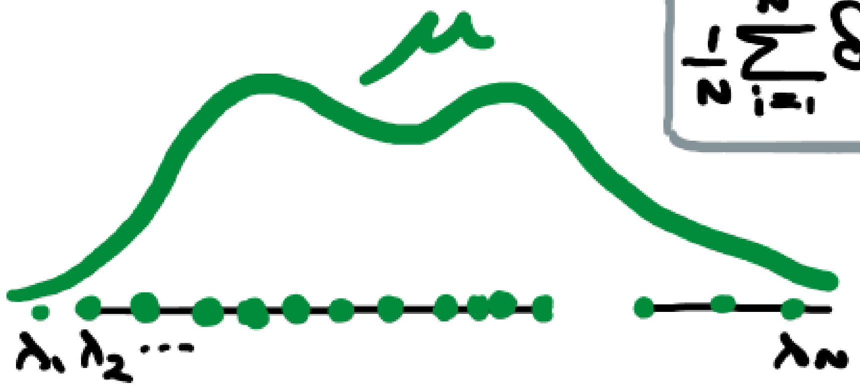

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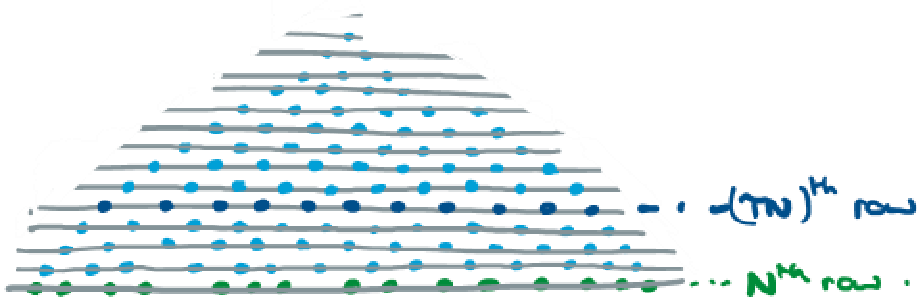
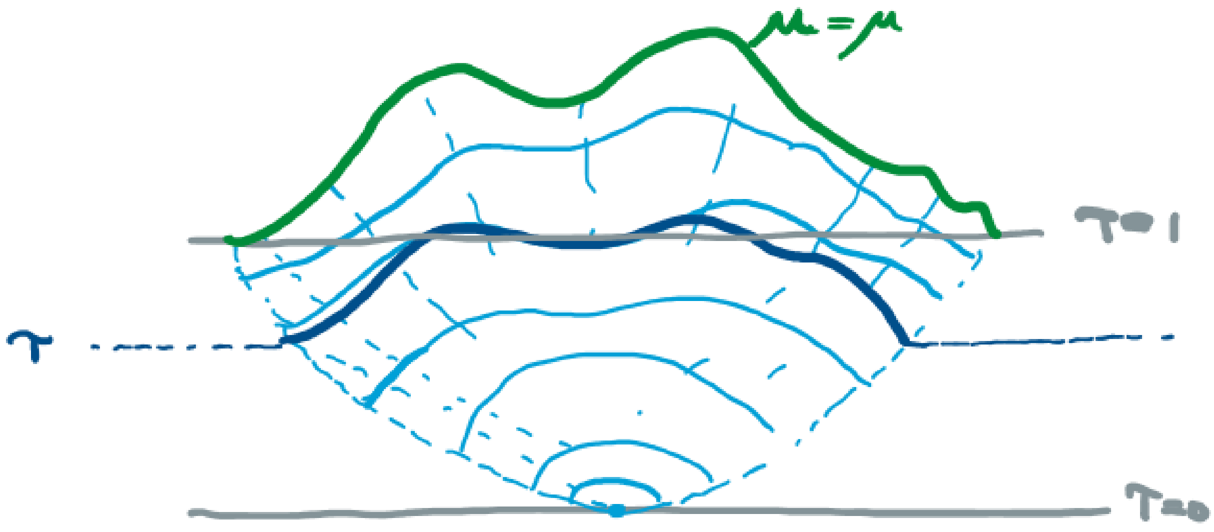
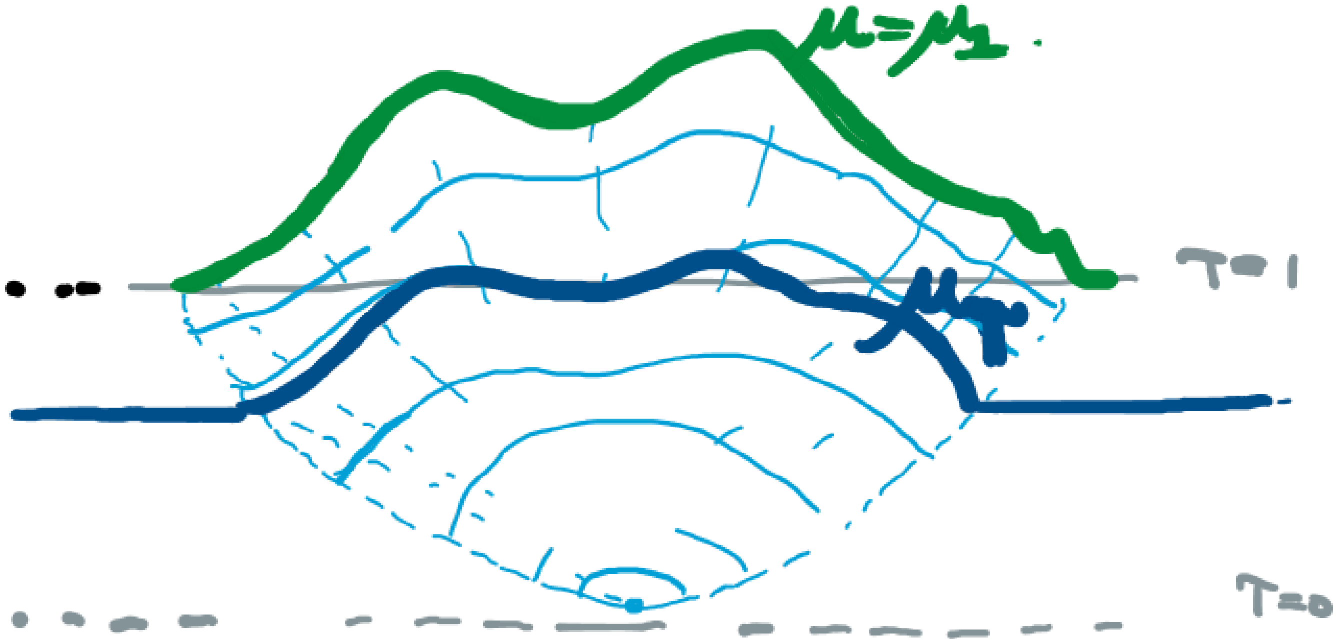
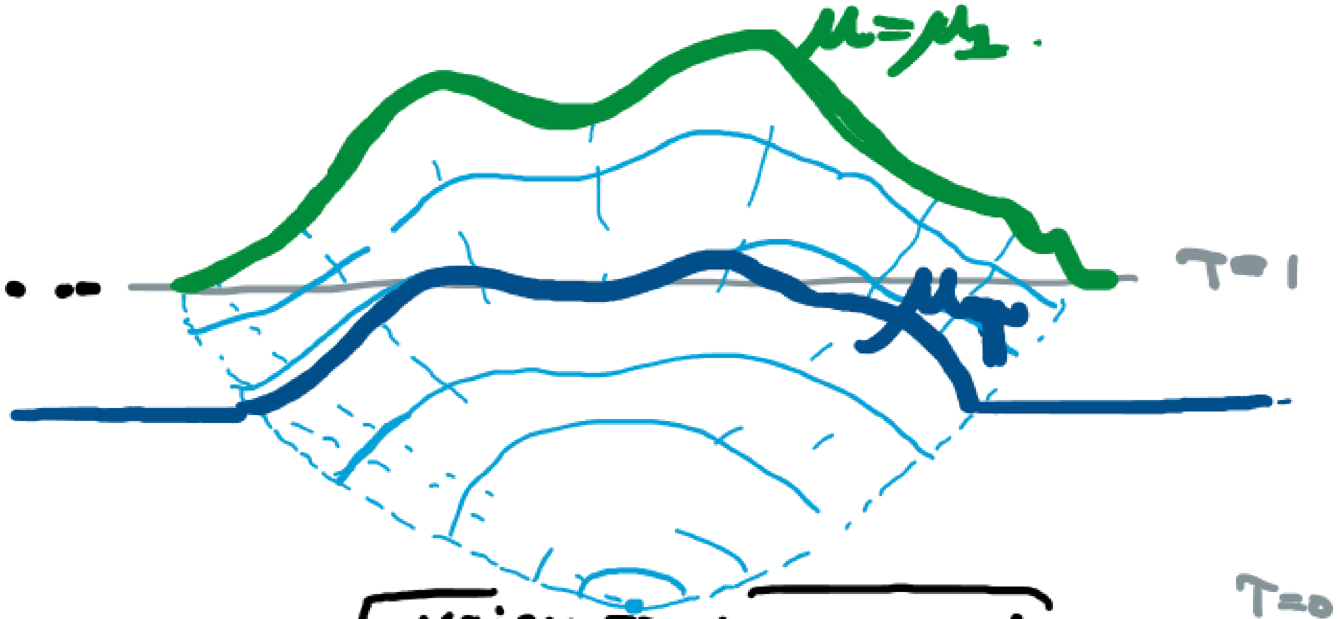


WHAT IS THE MACROSCOPIC SHAPE OF THE GIANT RANDOM PATTERN?



Let $\tau \in (0,1)$. $k_N = \lfloor \tau N \rfloor$





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THEOREM VOICULESCU, NICA-SPEICHER 90's

GIVEN μ, $\tau \in [0,1]$, THERE IS A MEASURE μ_τ SUCH THAT (INFORMALLY):

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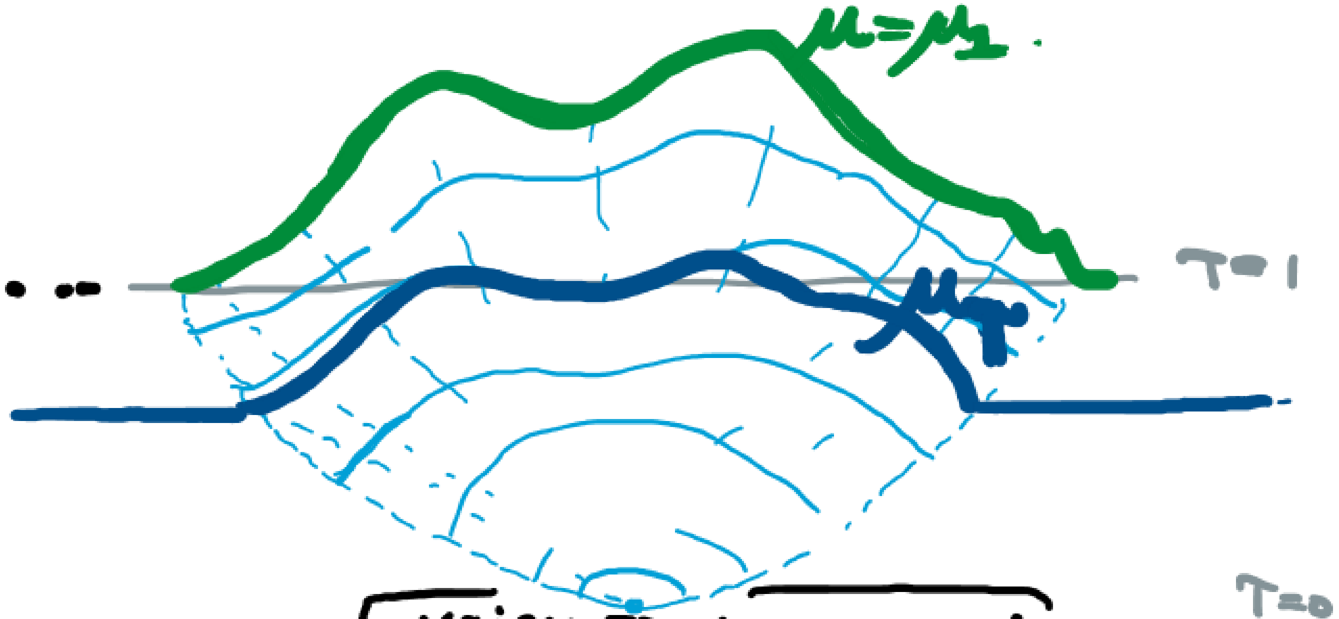


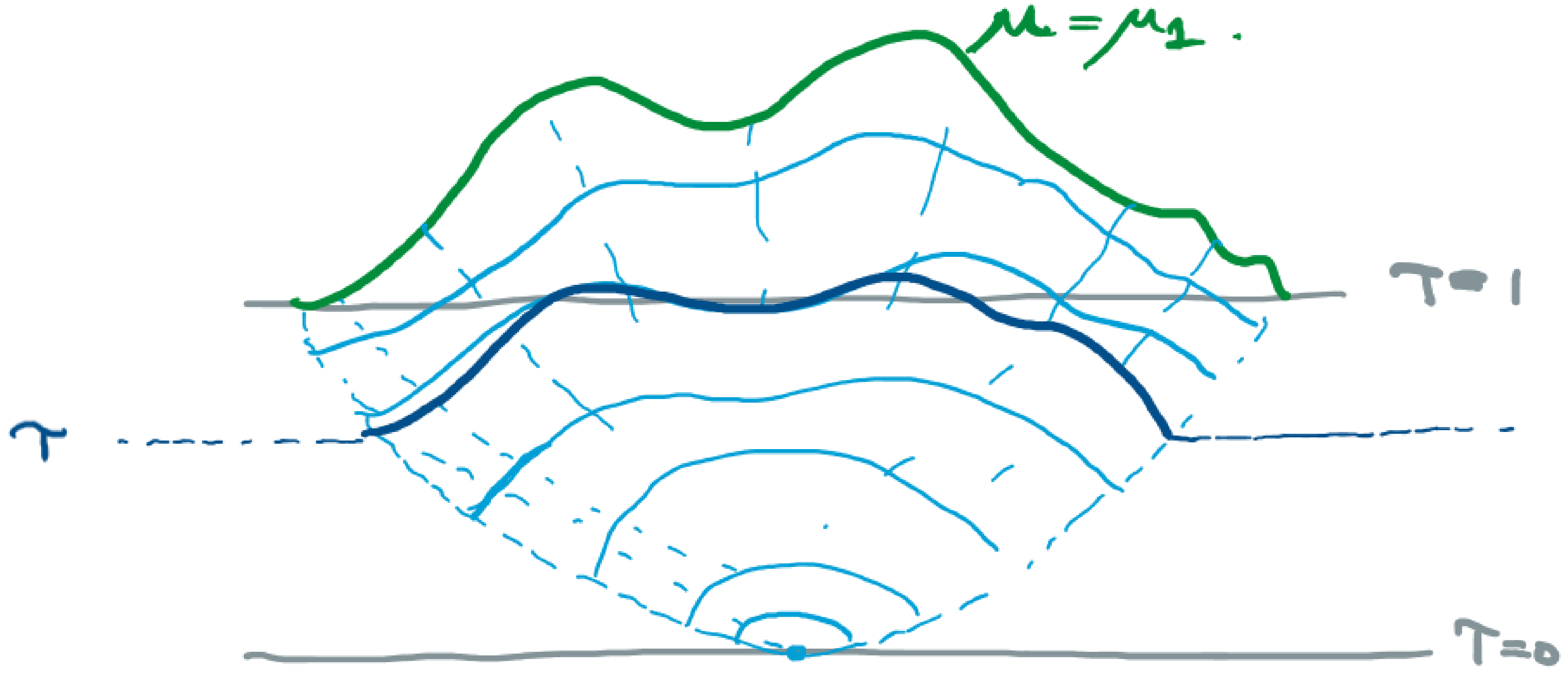
THEOREM VOICULESCU, NICA-SPEICHER 90's

GIVEN μ, $\tau \in [0,1]$, THERE IS A MEASURE μ_τ SUCH THAT (INFORMALLY):

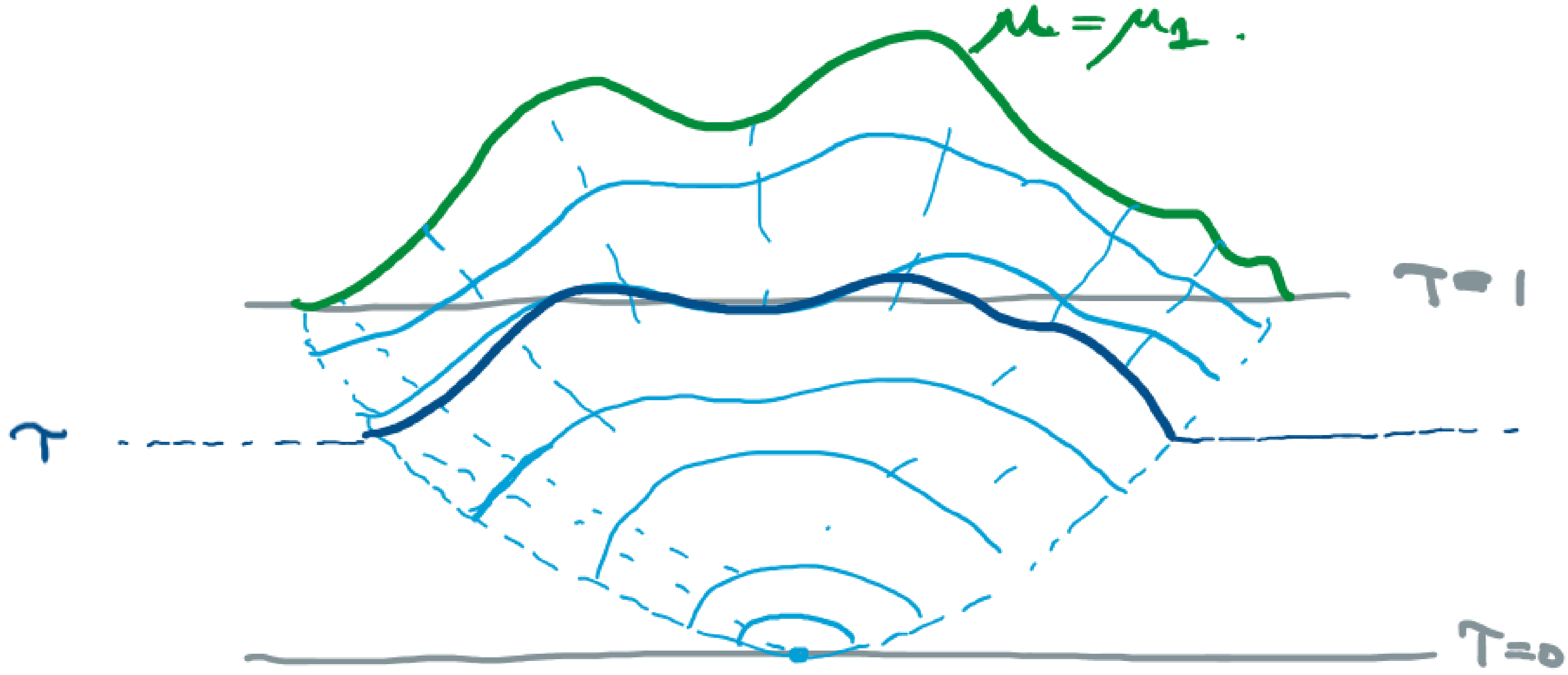
IF $\mu \approx$ EIGENVALUES OF RANDOM MATRIX A_N

THEN $\mu_\tau \approx$ EIGENVALUES OF $\lfloor \tau N \rfloor \times \lfloor \tau N \rfloor$ MINOR B_N (WITH HIGH PROB AS $N \to \infty$)

THE MEASURES $\{\mu_\tau : \tau \in [0,1]\}$ CAN BE FOUND BY LOOKING AT TRACES OF POWERS OF MINORS OF LARGE RANDOM MATRICES

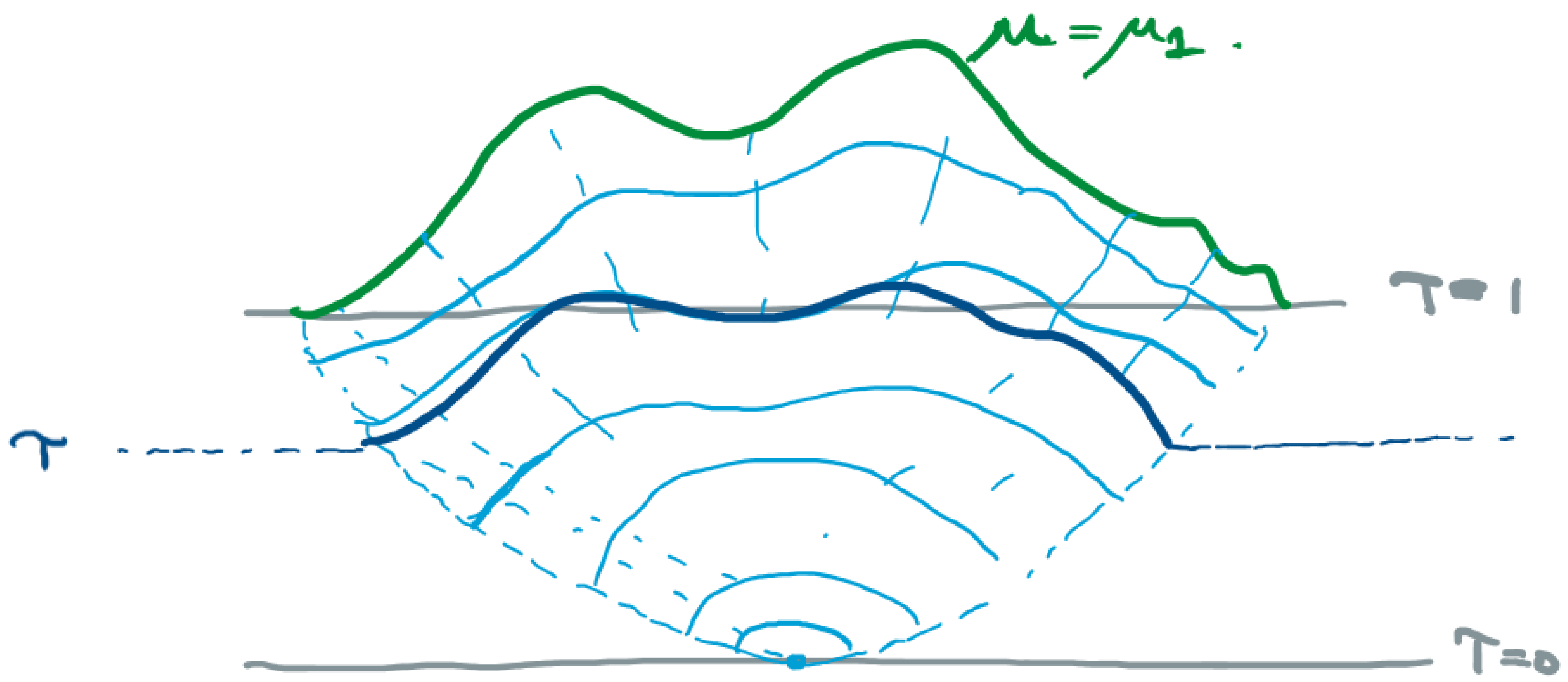


THE MEASURES $\{\mu_T : T \in [0,1]\}$ CAN BE FOUND BY LOOKING AT TRACES OF POWERS OF MINORS OF LARGE RANDOM MATRICES

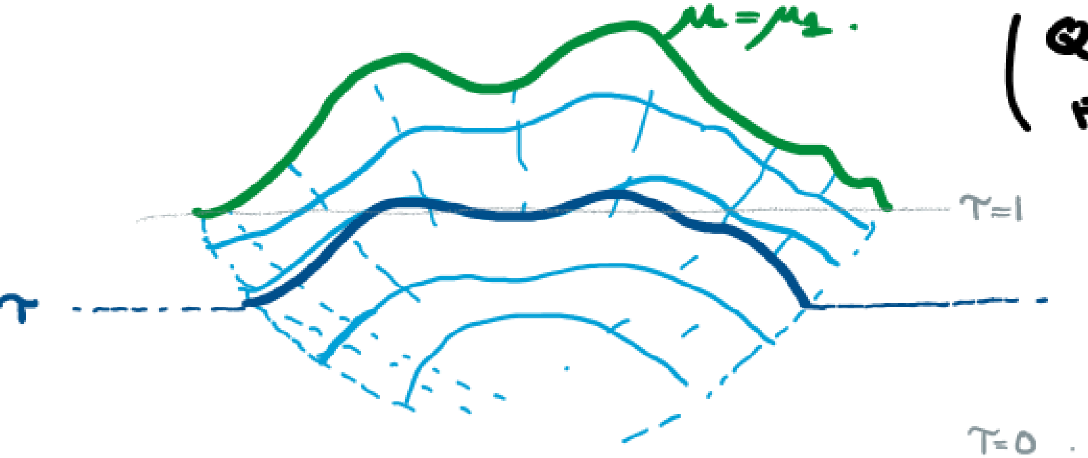


AND THEY CAN BE CHARACTERISED USING "CAUCHY TRANSFORMS"

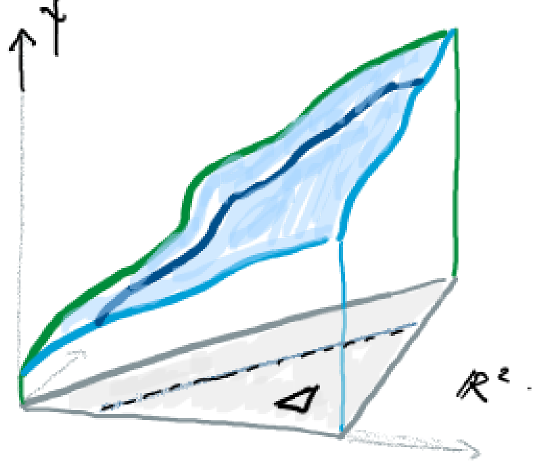
$$G_\mu(z) := \int_{-\infty}^{\infty} \frac{1}{z-x}\,\mu(dx) \qquad z \in \mathbb{C} - \mathbb{R}.$$



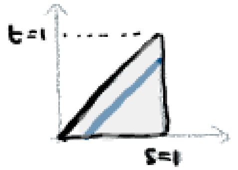
SHLYAKHTENKO + TAO (2022) RECENTLY OBSERVED THAT THE COLLECTION OF MEASURES $\{\mu_\tau : \tau \in [0,1]\}$ HAS A VARIATIONAL DESCRIPTION...

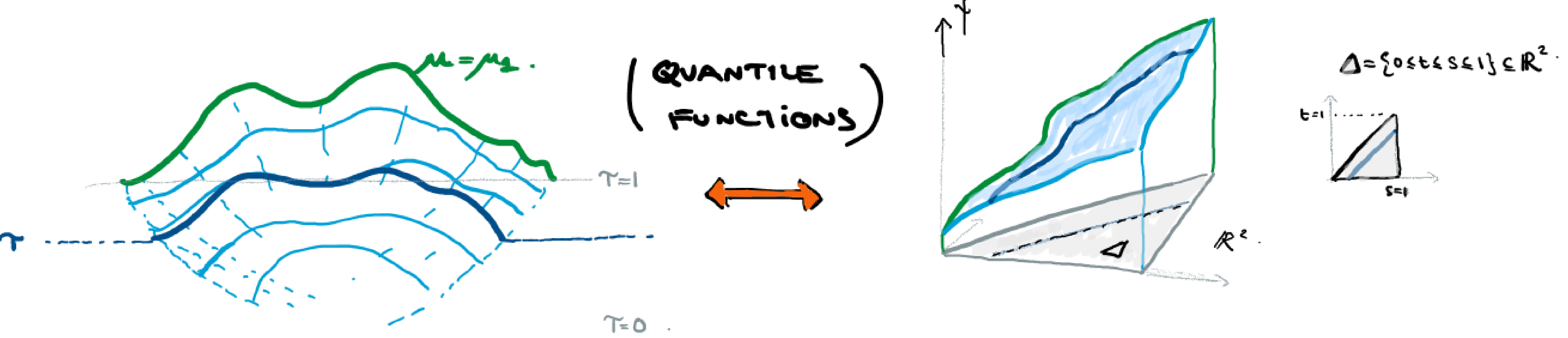


(QUANTILE FUNCTIONS)



$\Delta = \{0 \le t \le s \le 1\} \subseteq \mathbb{R}^2 .$

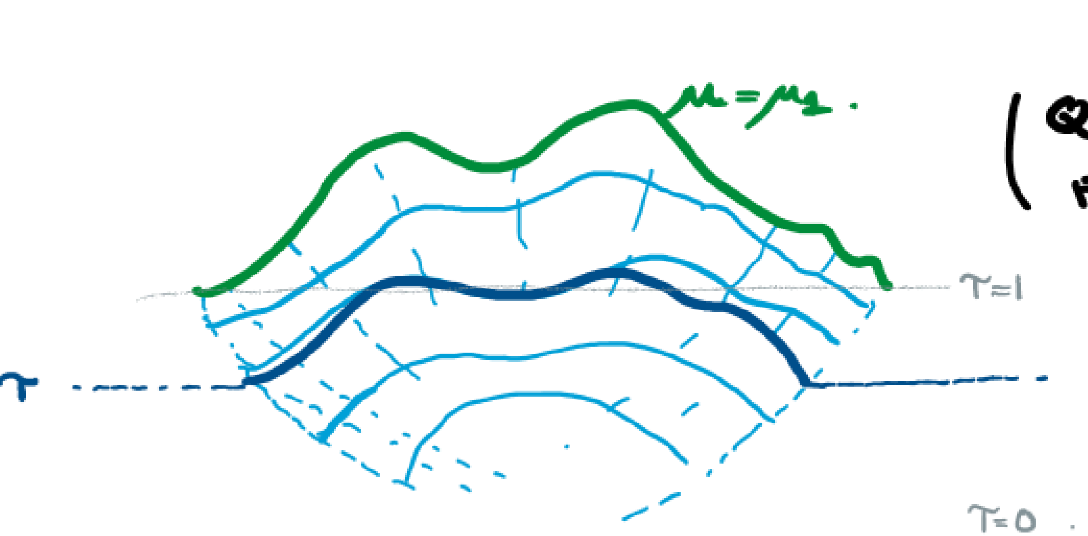




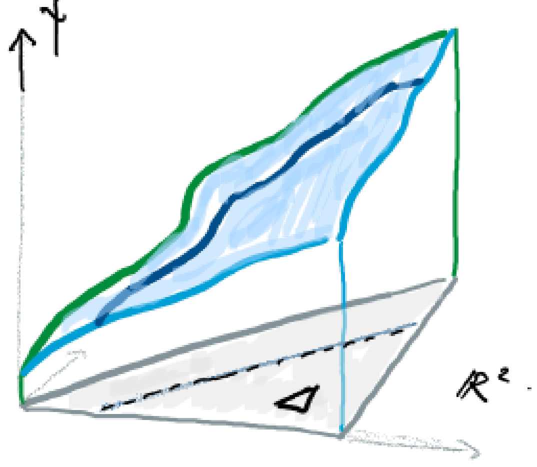
THEOREM (SHLYAKHTENKO + TAO 2020)

THE FUNCTION $\psi : \Delta \longrightarrow \mathbb{R}$ ASSOCIATED WITH FREE COMPRESSIONS $\{\mu_\tau : \tau \in [0,1]\}$ OF A MEASURE μ SATISFIES EULER-LAGRANGE EQUATIONS ASSOCIATED WITH MINIMISING

$$\mathcal{E}[\psi] := \int_\Delta \sigma(\nabla\psi)\, ds\, dt$$



(QUANTILE FUNCTIONS) $\longleftrightarrow$



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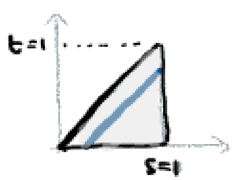

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$$\mathcal{E}[\psi] := \int_\Delta \sigma(\nabla\psi)\, ds\, dt$$

WHERE

$$\sigma(\nabla\psi) = -\log(\psi_s + \psi_t) - \log\sin\left(\pi \frac{\psi_s}{\psi_s + \psi_t}\right)$$

CONJECTURE (SHLYAKHTENKO + TAO 2020)

THE FUNCTION

$$\sigma_0(u_1, u_2) = -\log(u_1+u_2) - \log\sin\pi\frac{u_1}{u_1+u_2}$$

is (UP TO CONSTANTS) "THE SURFACE TENSION OF THE BEAD MODEL"

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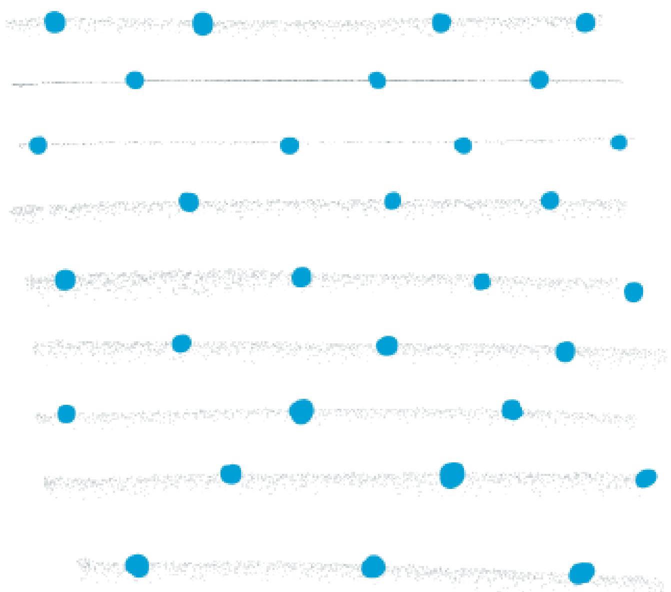


BOUTILLIER'S (2009) BEAD MODEL:

"BEADS" ON STRINGS AT UNIT DENSITY

RULE: BETWEEN EVERY TWO CONSECUTIVE BEADS ON A STRING

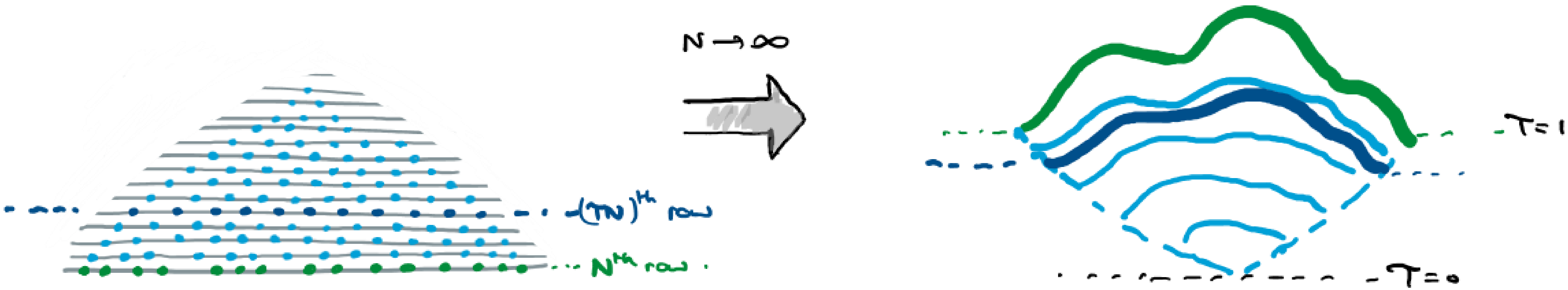
THERE ARE BEADS ON NEIGHBOURING STRINGS

2. Goal

Our goal is to study Gelfand-Tsetlin Patterns from a purely statistical-
-physics perspective.

FORGET ABOUT EIGENVALUES AND MATRICES!

LETS MAKE THIS ALL ABOUT THE STATISTICAL PHYSICS BEHAVIOUR OF LARGE UNIFORM GT-PATTERNS.



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LET μ BE PROB MEASURE. FOR LARGE N, LET $\lambda_1 \leq \cdots \leq \lambda_N$ CHOSEN SO THAT

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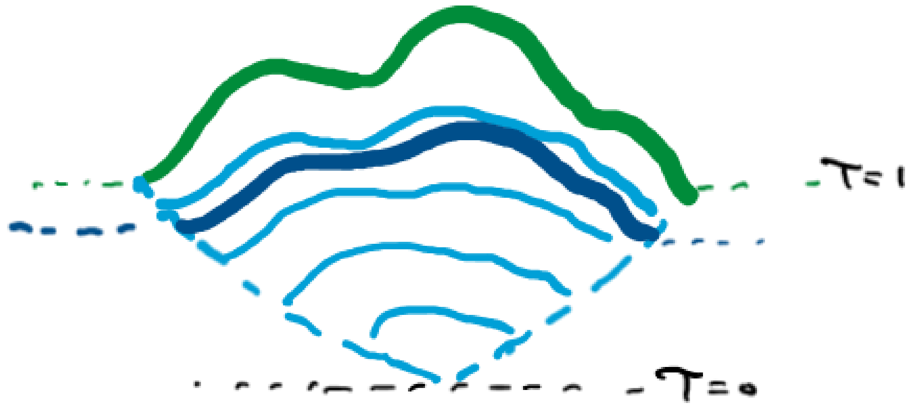
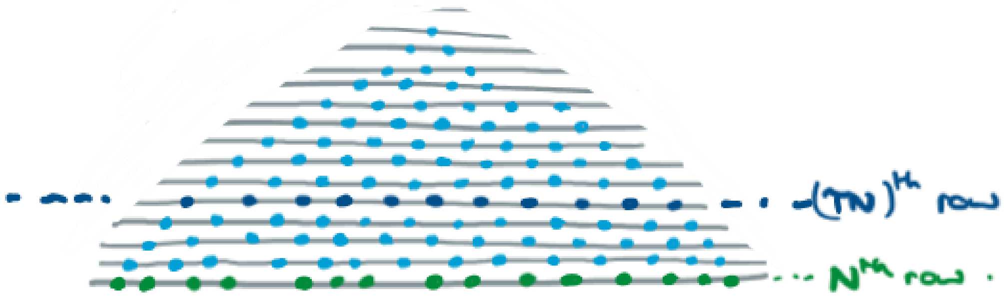


- CHOOSE A UNIFORM GT-PATTERN WITH BOTTOM ROW $\lambda_1 \leq \cdots \leq \lambda_N$



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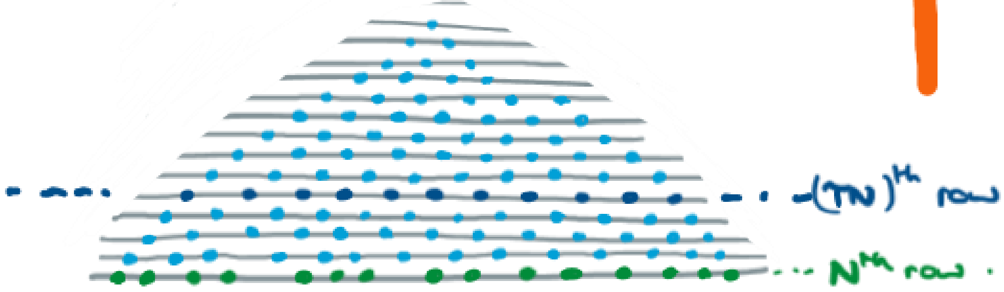
GOAL

CAN WE SHOW $(TN)^{th}$ ROW CONCENTRATES AROUND μ_T AS $N \to \infty$?



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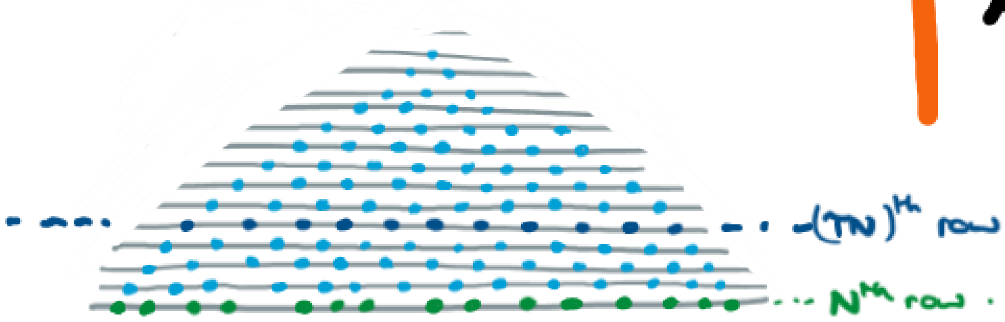
- CHOOSE A UNIFORM GT-PATTERN WITH BOTTOM ROW $\lambda_1 \leq \cdots \leq \lambda_N$

GOAL

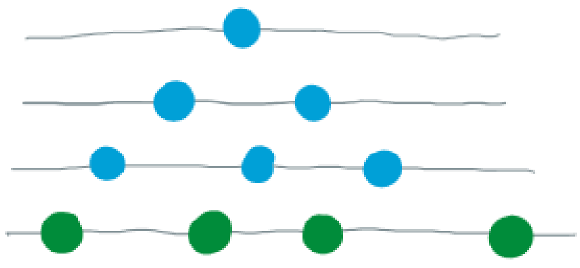
CAN WE SHOW $(TN)^{th}$ ROW CONCENTRATES AROUND μ_T AS $N \to \infty$?

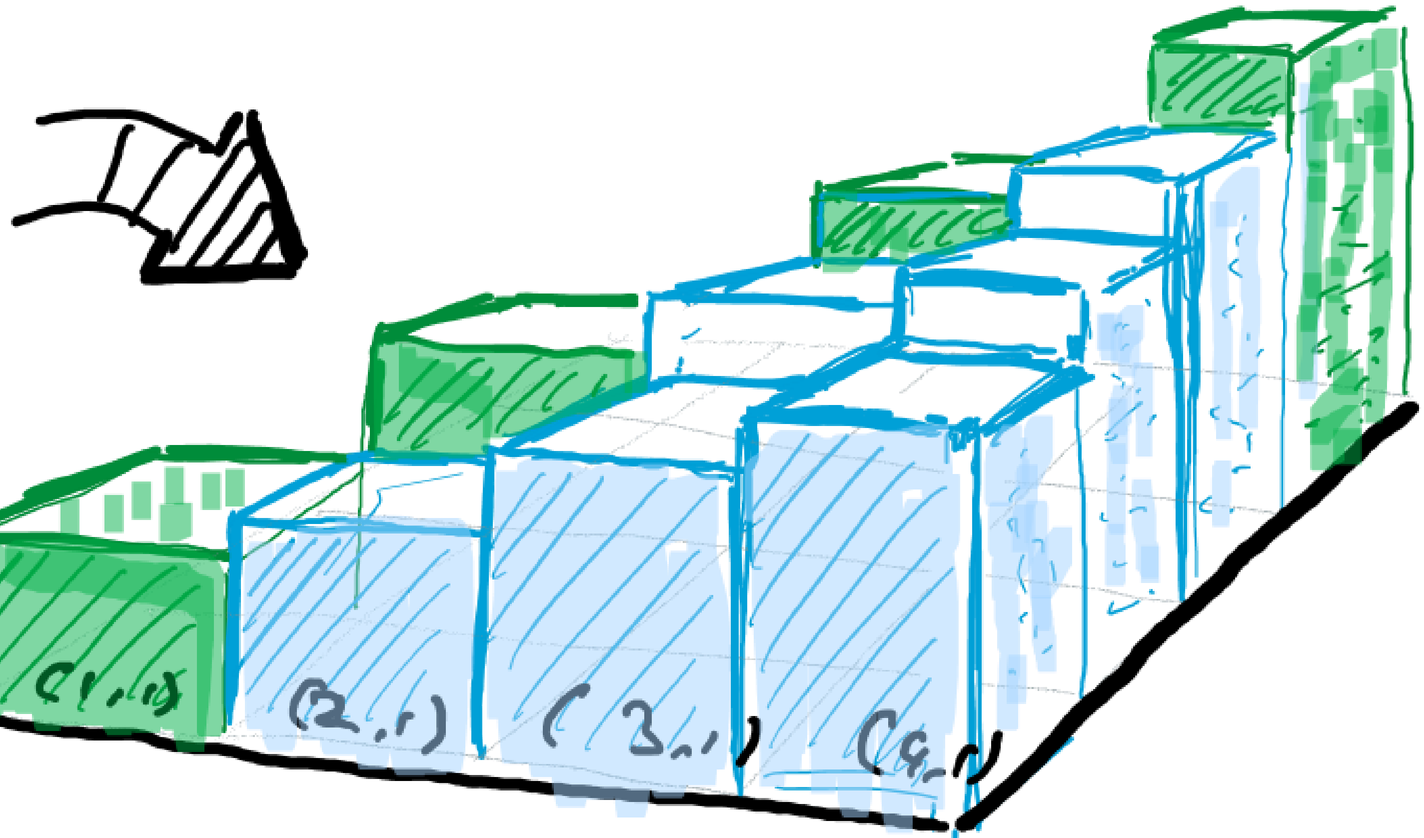
- DOES OUR ANSWER EXPLAIN SHLYAKHTENKO & TAO'S VARIATIONAL CHARACTERISATION?



3. Random Gelfand-Tsetlin Patterns as random Surfaces.

OBSERVATION:

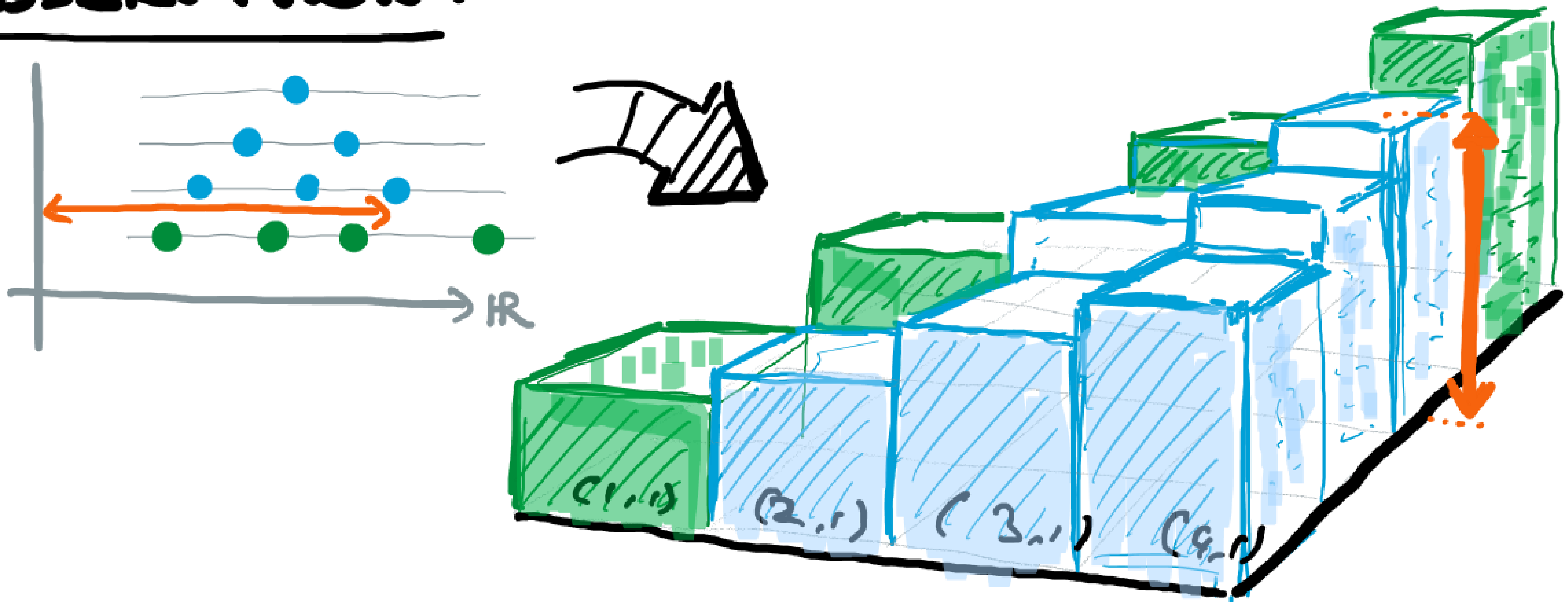




A GELFAND–TSETLIN PATTERN IS THE SAME AS A FUNCTION $\phi : \Delta_n \longrightarrow \mathbb{R}$ $\quad \Delta_n = \{(x_1, x_2) \in \mathbb{Z}^2 : 1 \leq x_2 \leq x_1 \leq N\}$

THAT INCREASES IN BOTH COORDINATES.

OBSERVATION :



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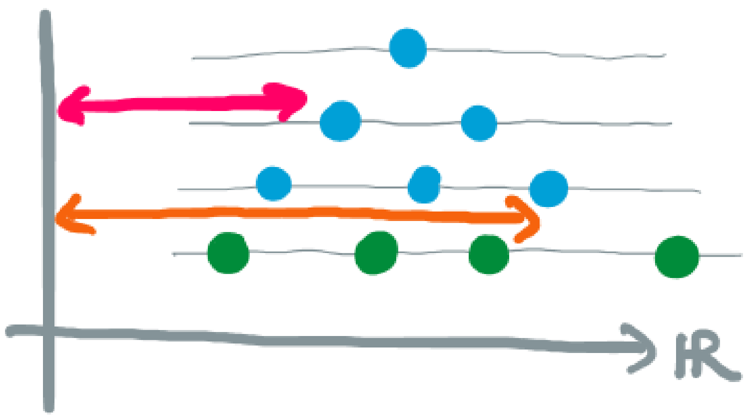

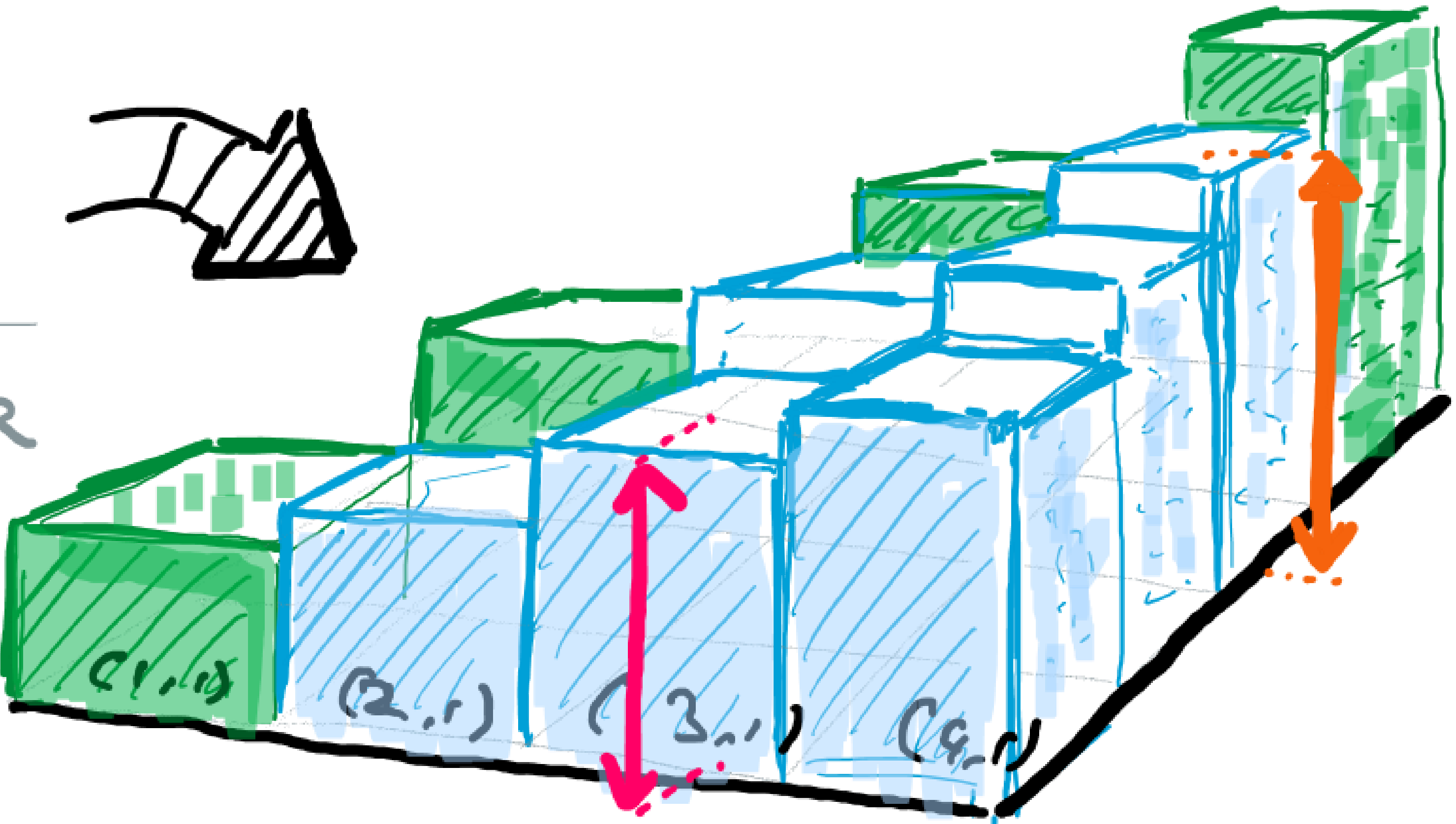

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A GELFAND-TSETLIN PATTERN IS THE SAME AS A FUNCTION $\phi : \Delta_n \longrightarrow \mathbb{R}$ $\quad \Delta_n = \{(x_1, x_2) \in \mathbb{Z}^2 : 1 \leq x_2 \leq x_1 \leq n\}$

THAT INCREASES IN BOTH COORDINATES.

A RANDOM SURFACE (AKA GINZBURG-LANDAU $\nabla\phi$ INTERFACE MODEL) IS A RANDOM FUNCTION $\phi : C \to \mathbb{R}$ WITH DENSITY PROPORTIONAL TO

$C \subseteq \mathbb{Z}^2$

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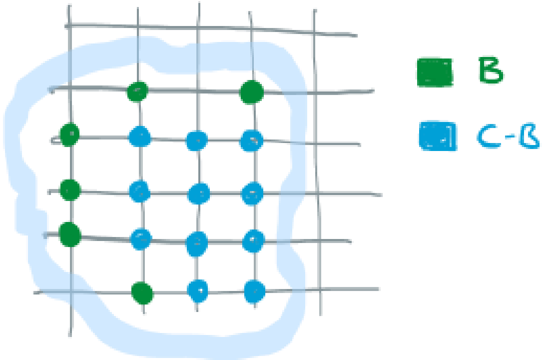

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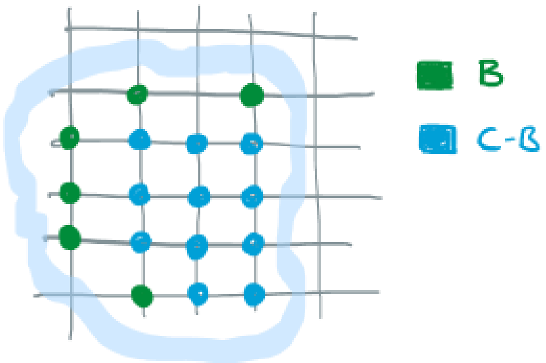

① SUM OVER ALL DIRECTED EDGES $x \to y$ $\begin{smallmatrix} y \\ \uparrow \\ x \end{smallmatrix}$ IN C.
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$$\exp\underbrace{\left\{-\sum_{x\to y \text{ in } C} V(\phi_y-\phi_x)\right\}}_{(1)}\underbrace{\prod_{x\in B}\delta_{\varphi_x}(d\phi_x)}_{(2)}\prod_{x\in C-B} d\phi_x$$

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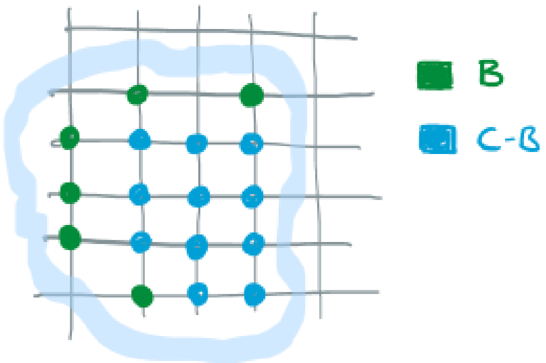

(1) SUM OVER ALL DIRECTED EDGES $x \to y$, $\underset{x}{\overset{y}{\uparrow}}$ IN C. $V : \mathbb{R} \to \mathbb{R}$ IS AN "INTERACTION POTENTIAL".

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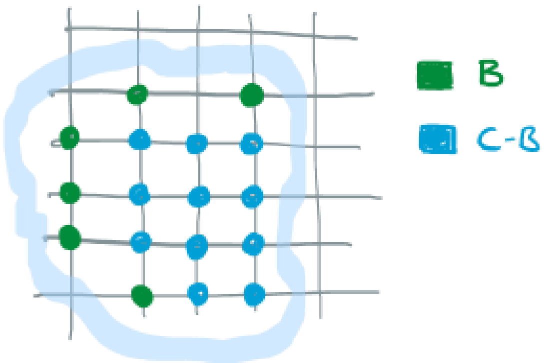

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3 ϕ_x a.c. WRT Lebesgue MEASURE $x \in C-B$.

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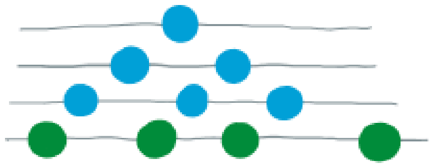

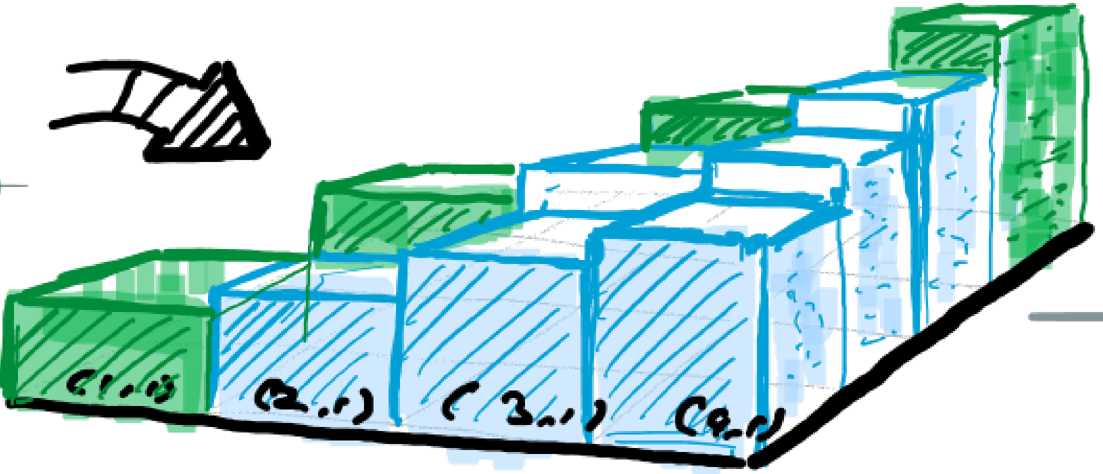

THE DENSITY OF THIS $\phi : \Delta_n \to \mathbb{R}$ CAN BE WRITTEN.

$$\propto \exp\left\{ - \sum_{x \sim y \text{ in } C} V(\phi_y - \phi_x) \right\} \prod_{x \in B} \delta_{\varphi_x}(d\phi_x) \prod_{x \in C - B} d\phi_x$$

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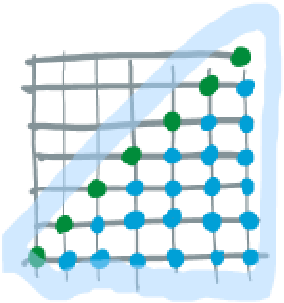

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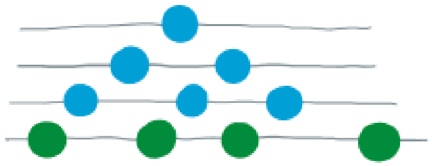

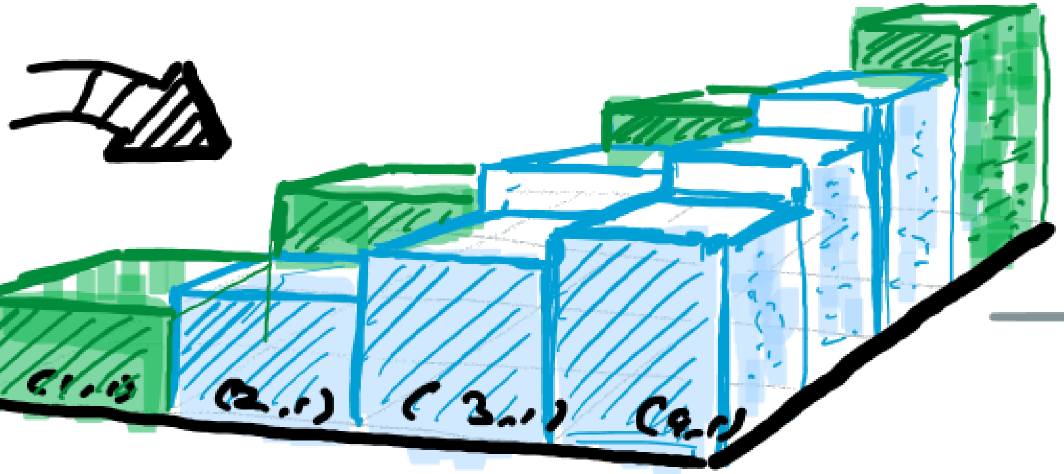

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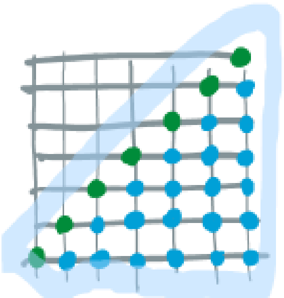

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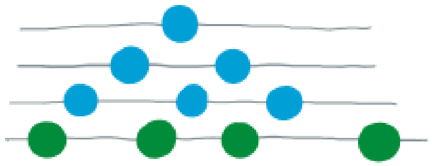

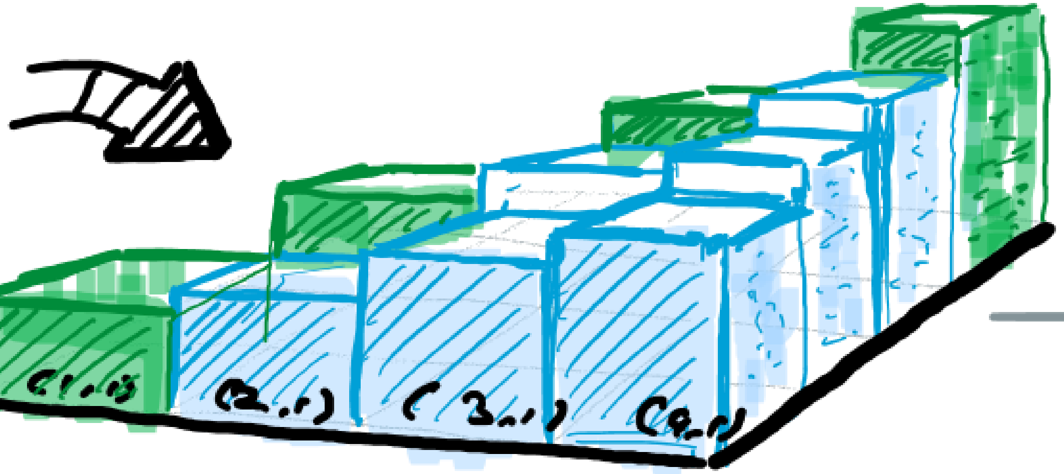

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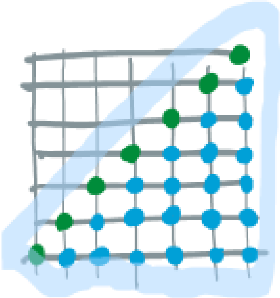

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HARDCORE INTERACTION

$$V(\phi_y-\phi_x)=\begin{cases} 0 & \text{if } \phi_y\geq\phi_x \\ +\infty & \text{if } \phi_y<\phi_x \end{cases}$$

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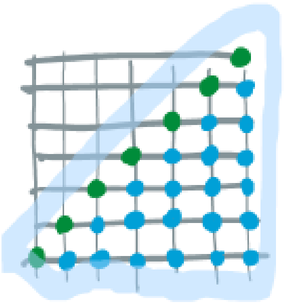

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NOTE IN THIS CASE

$$\exp\left\{ - \sum_{x,y} V(\phi_y - \phi_x) \right\} = \mathbb{1}\left\{ \phi \text{ is increasing in both coordinates} \right\}$$

THUS GELFAND-TSETLIN PATTERNS ARE A SPECIAL CASE OF A RANDOM SURFACE $\phi : C \longrightarrow \mathbb{R}$

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- ONE CAN DEFINE RANDOM SURFACES IN $\mathbb{Z}^d$.
- THE CASE $V(u) = au^2$ IS THE DISCRETE GAUSSIAN FREE FIELD
- IN 99% OF LITERATURE

$$\boxed{\text{ASSUMPTION}: \quad 0 < c_1 \leq V''(u) \leq c_2 < \infty}$$

THERE IS A RICH LITERATURE ON RANDOM SURFACES UNDER THE $0 < c_1 \leq V''(u) \leq c_2 < \infty$ ASSUMPTION

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WARNING

OUR INTERACTION POTENTIAL $V(u) = \infty \mathbb{1}\{u<0\}$ IS NOT EVEN CONTINUOUS OR FINITE...

... LET ALONE UNIFORMLY TWICE DIFFERENTIABLE .

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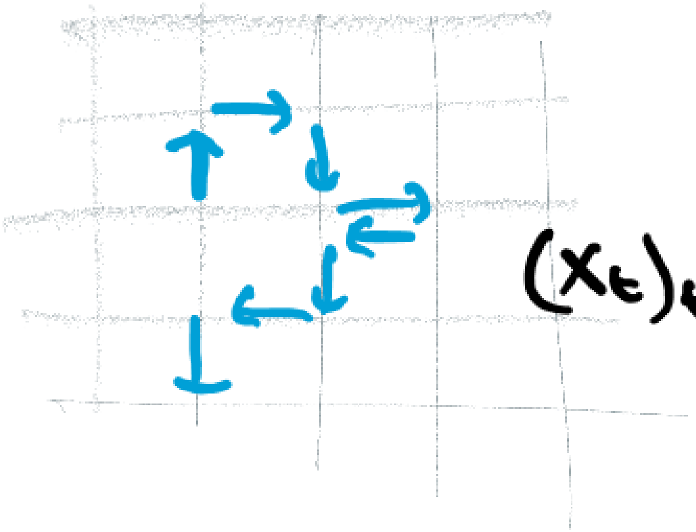
LETS LOOK AT SOME OF THIS LITERATURE ANYWAY.

UNDER $0<c_1\le V''(u)\le c_2<\infty$ ASSUMPTION,

THE HELFFER-SJÖSTRAND (1994) REPRESENTATION RELATES COVARIANCES TO RANDOM WALK IN RANDOM ENVIRONMENT:

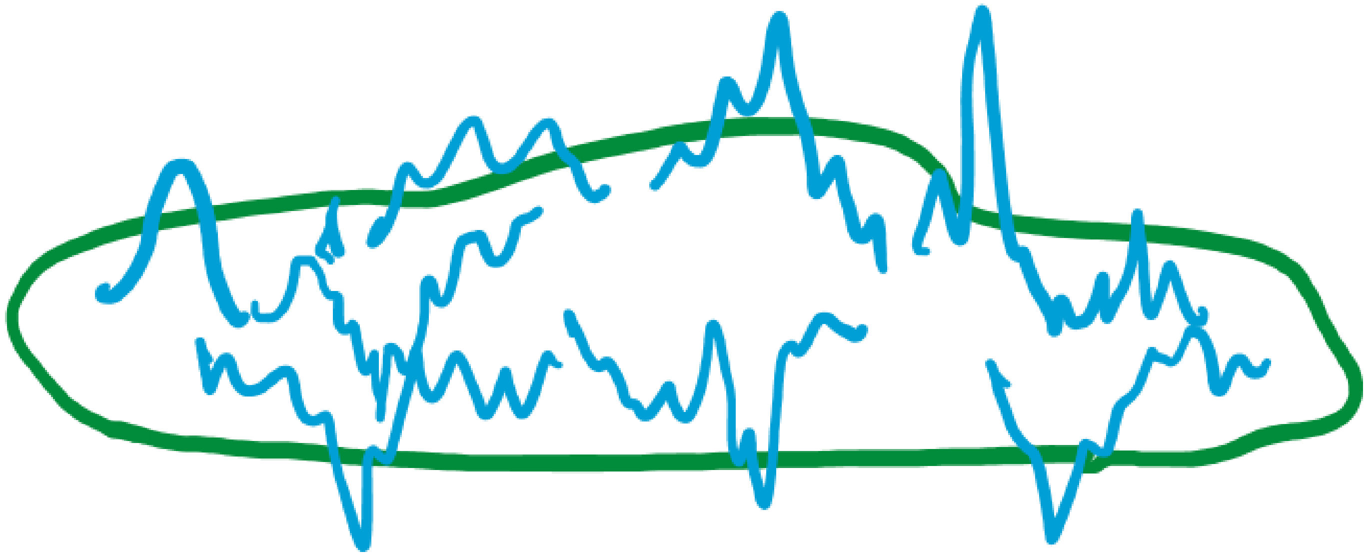
$$\mathbb{E}[\phi_x\phi_y] = \mathbb{E}_x\left[\int_0^\infty \mathbb{1}\{X_t=y\}\,dt\right]$$



$(X_t)_{t\geq 0}$

UNDER $0 < c_1 \leq V''(u) \leq c_2 < \infty$ ASSUMPTION,

RESCALED SURFACES HAVE GAUSSIAN FREE FIELD FLUCTUATIONS. (MILLER 2011).



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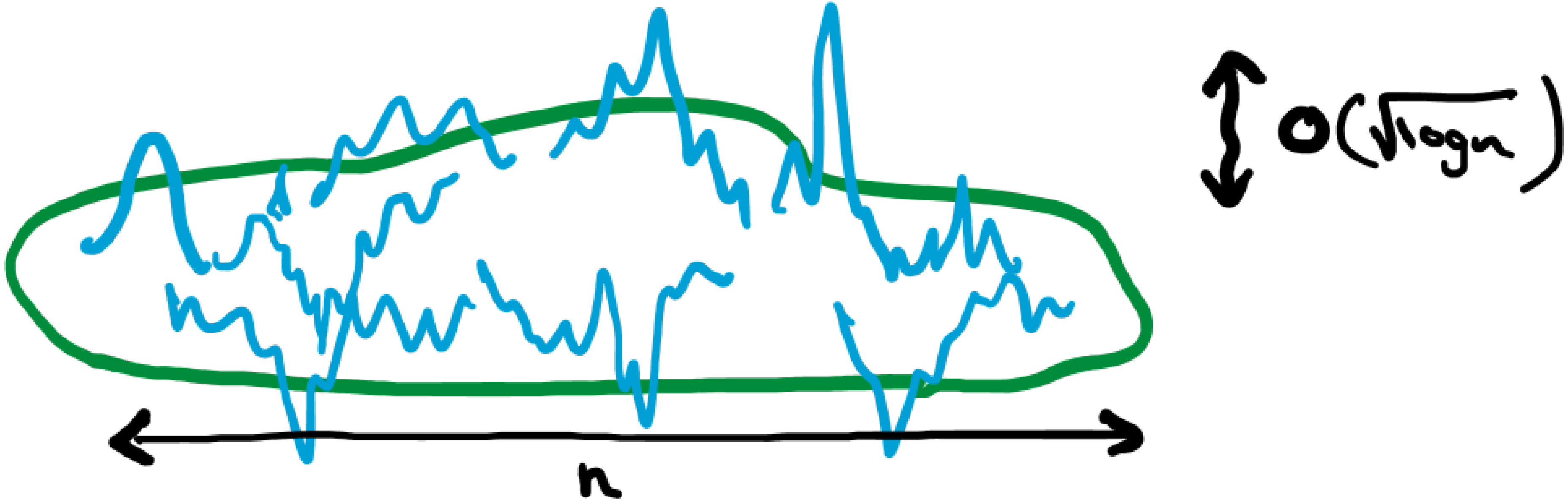
(IN TWO DIMENSIONS)

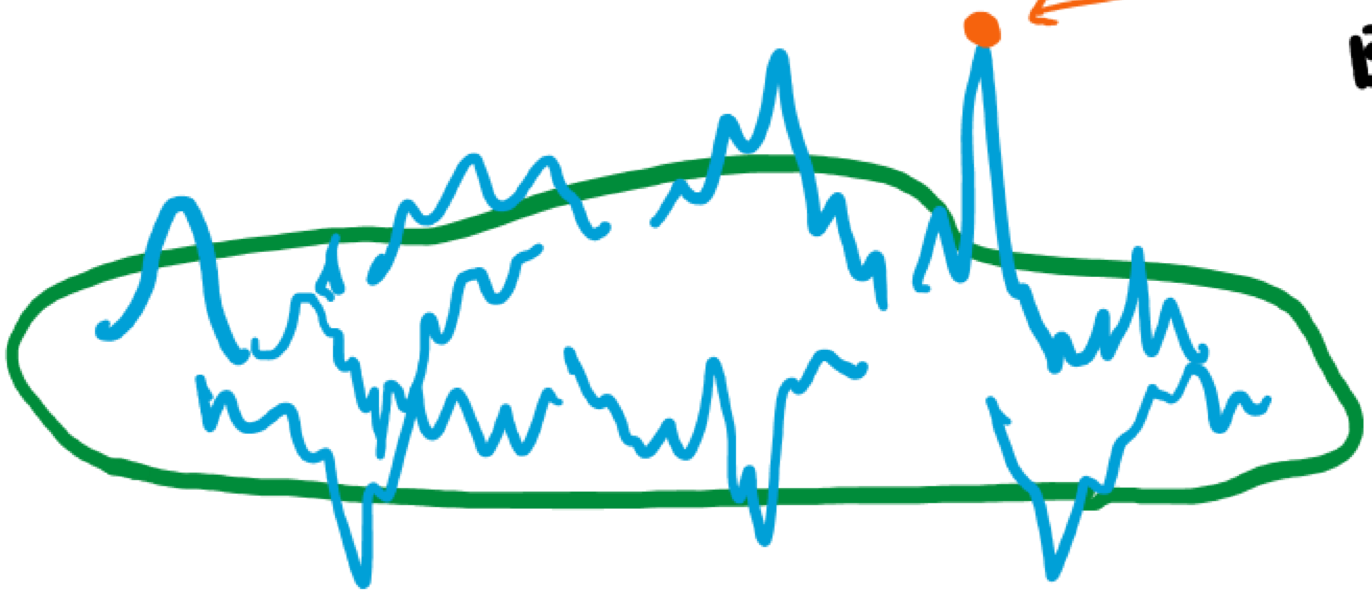


VARIANCES ARE $O(\log n)$.

UNDER $0 < c_1 \le V''(u) \le c_2 < \infty$ ASSUMPTION,

(IN TWO-DIMENSIONS)

THE MAXIMA HAVE BEHAVIOUR RELATED TO BRANCHING RANDOM WALK.



BOLTHAUSEN - DEUSCHEL - GIACOMIN (2001)

DAVIAUD (2006)

BRAMSON - DING - ZEITOUNI (2016)

BELIUS - WU (2020).

MOST IMPORTANTLY FOR US HOWEVER...

UNDER $0 < c_1 \leq V''(u) \leq c_2$ ASSUMPTION, FOR EACH POTENTIAL V THERE EXISTS A FUNCTION CALLED THE "SURFACE TENSION"

$\sigma_V : \mathbb{R}^2 \to \mathbb{R}$

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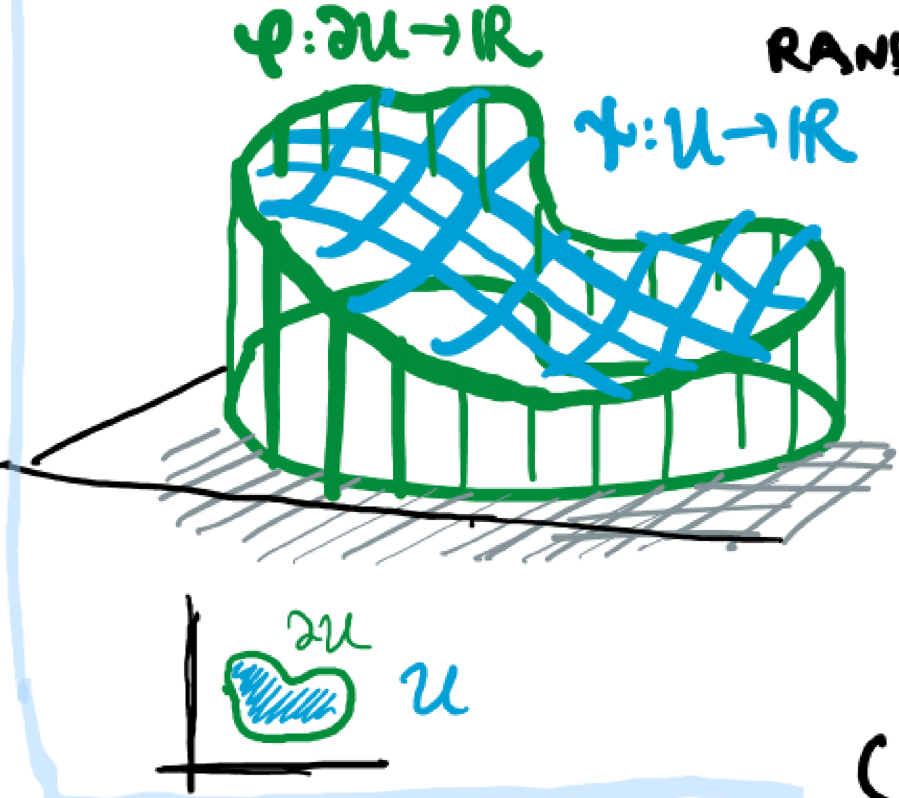
THEOREM DEUSCHEL-GIACOMIN-IOFFE (2000) FUNAKI-NISHIKAWA (2001).



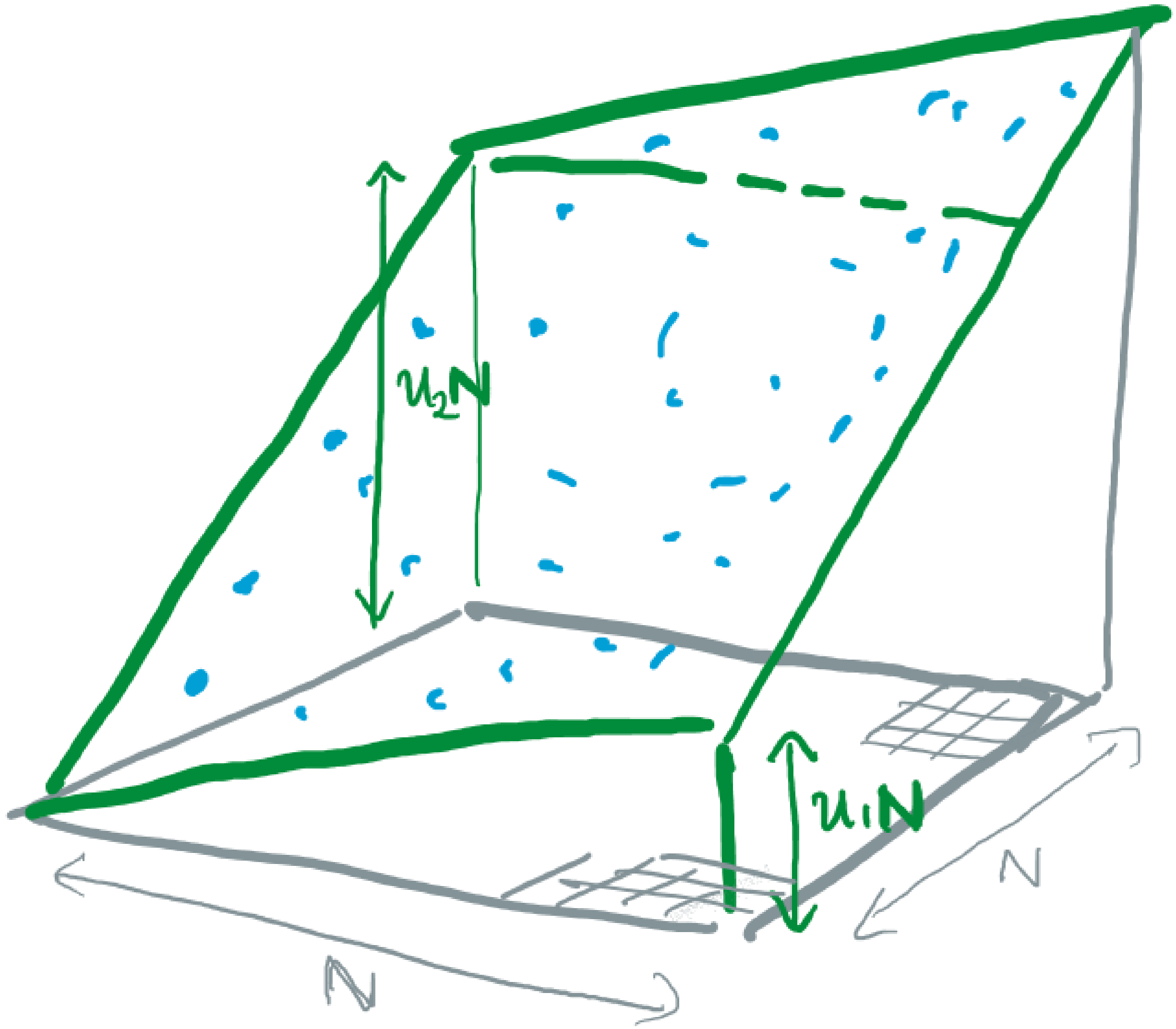
ON MACROSCOPIC SCALE, THE RESCALED RANDOM SURFACE $\psi : \mathcal{U} \to \mathbb{R}$ TRIES TO MINIMISE A SURFACE TENSION INTEGRAL

$$E[\psi] = \int_{\mathcal{U}} \sigma(\nabla\psi)\, ds\, dt .$$

SUBJECT TO $\psi = \varphi$ ON $\partial\mathcal{U}$.

(+ L.D.P.)

THE SURFACE TENSION DESCRIBES THE COST OF A LARGE PATCH OF INTERFACE TO LIE AT TILT $(u_1, u_2) \in \mathbb{R}^2$.



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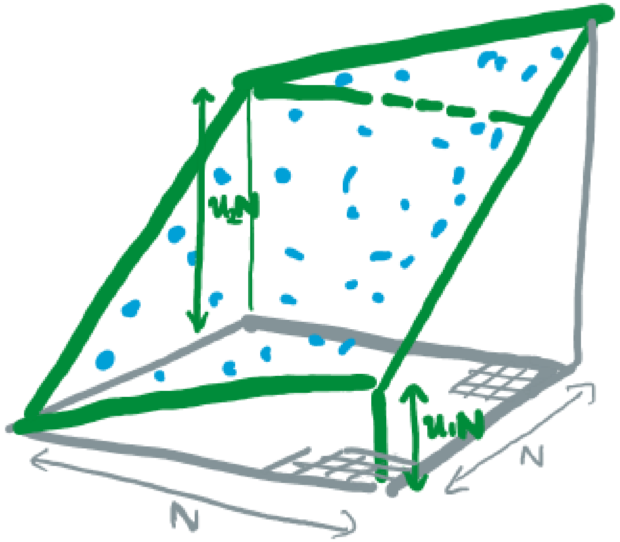

$$\sigma_V(u_1, u_2) := -\lim_{N \uparrow \infty} \frac{1}{N^2} \log Z_N(u_1, u_2)$$

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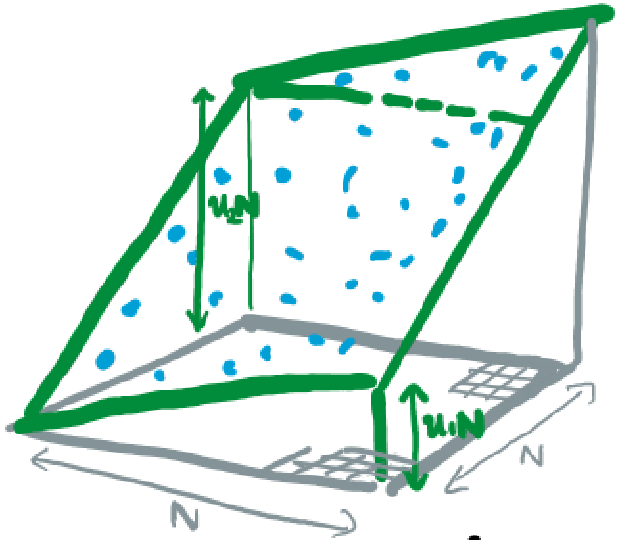

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LET $\square_N = \{0,1,\dots,N\}^2$, $\partial\square_N$ boundary, $\square_N^\circ$ INTERIOR.

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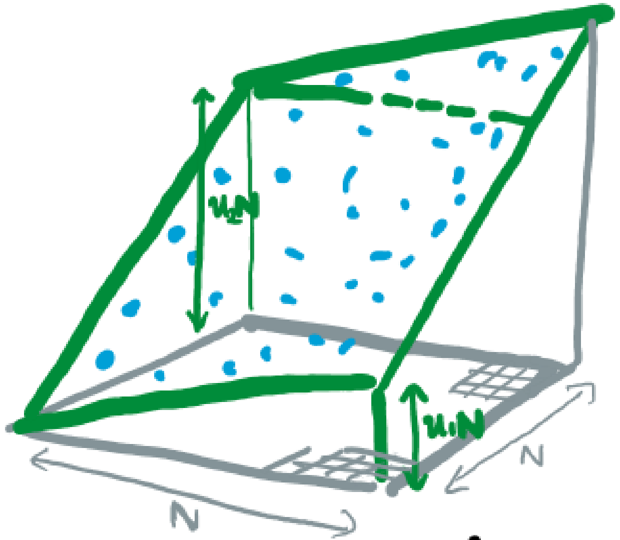

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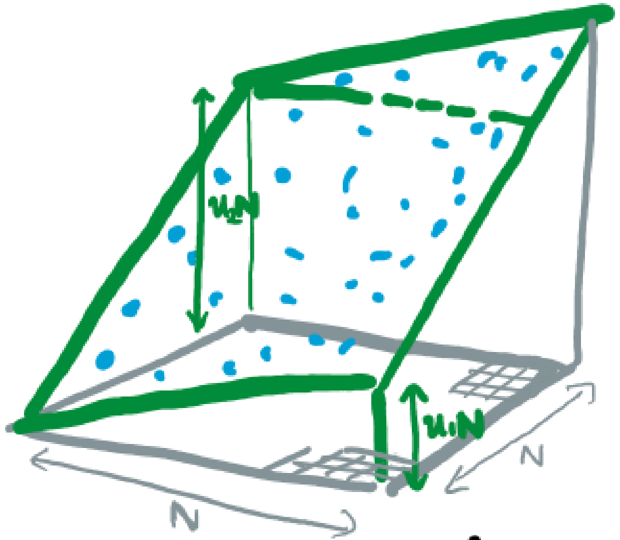

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BOUNDARY CONDITIONS FORCE $\phi_x = u_1 x_1 + u_2 x_2$

$x = (x_1, x_2) \in \partial \square_N$

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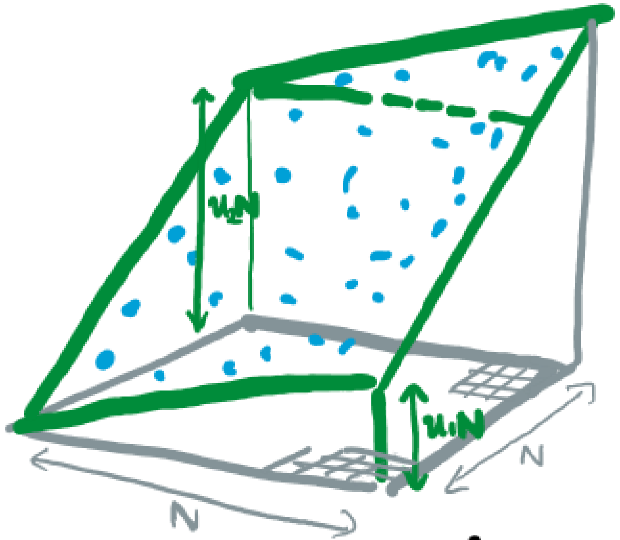

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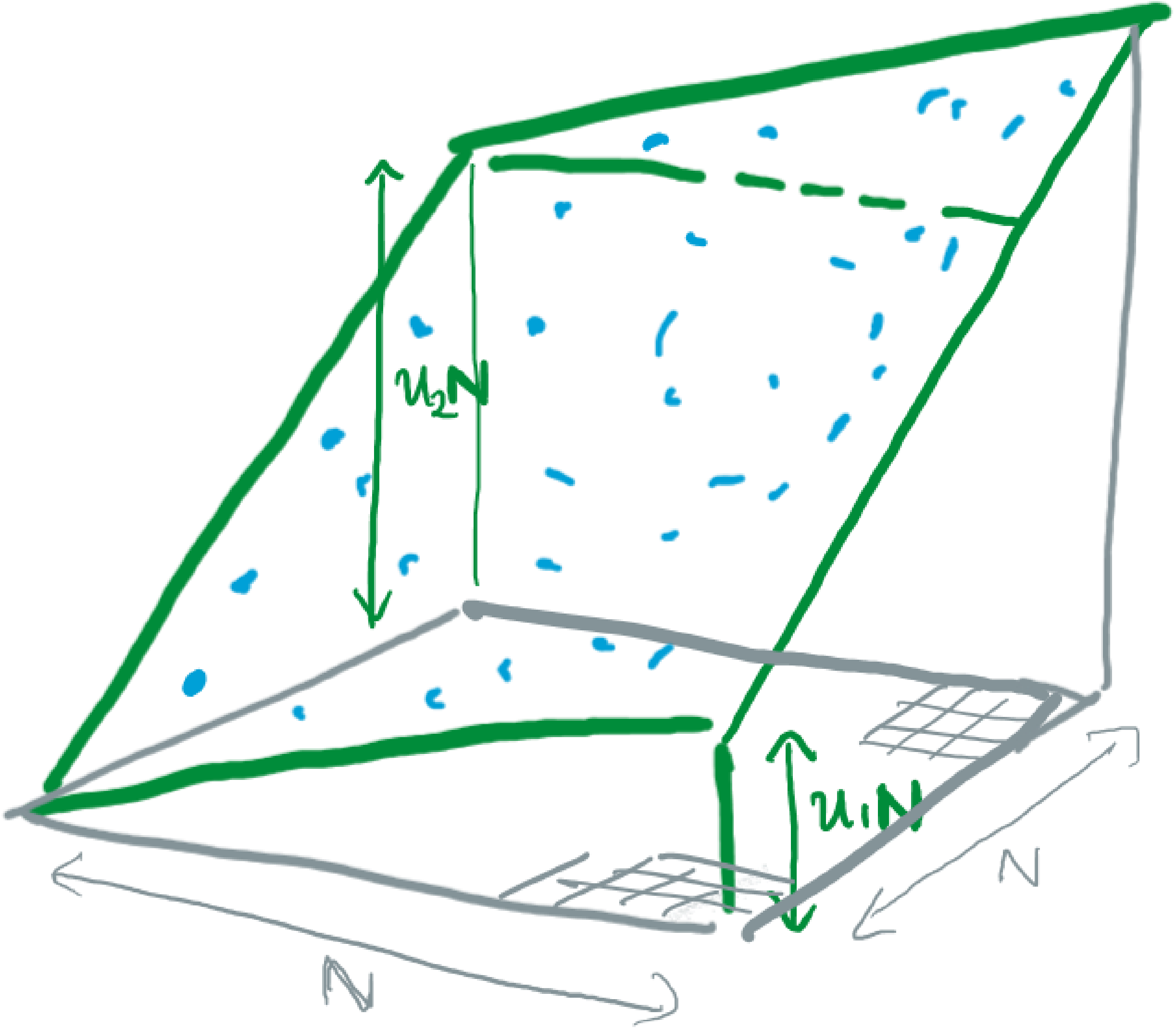
FUNAKI + SPOHN (1997): THIS LIMIT EXISTS WHEN $0 < c_1 \leq V''(u) \leq c_2$.

3. Main Results



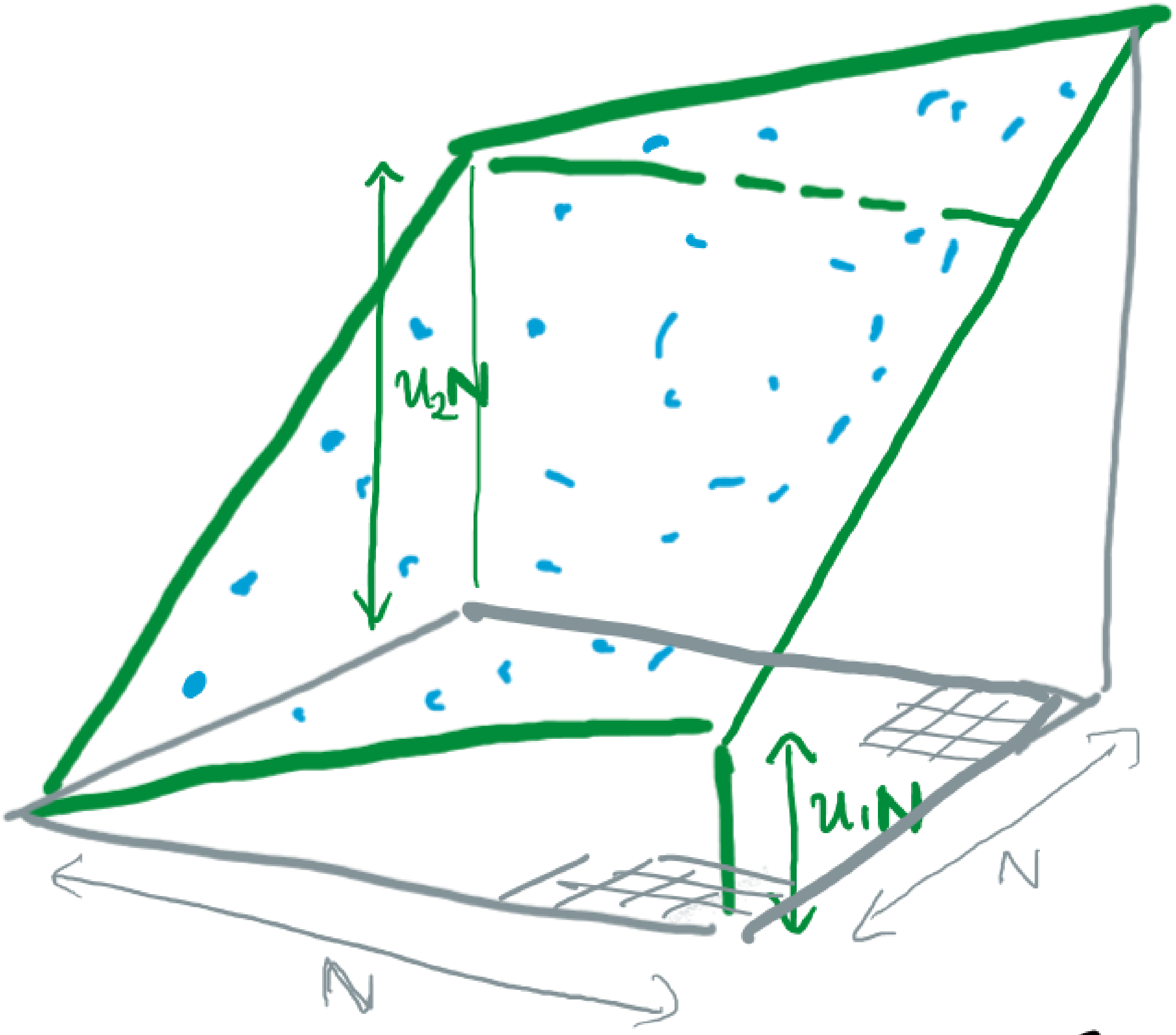
WHILE FUNAKI-SPOHN (1997) SHOWED THAT THE SURFACE TENSION

$$\sigma_V(u_1, u_2) := -\lim_{N\uparrow\infty} \frac{1}{N^2} \log Z_N(u_1, u_2)$$

EXISTS WHEN $0 < c_1 \leq V''(u) \leq c_2 < \infty$,

WE ONLY HAVE FORMULAS IN TWO SPECIAL CASES:

- One dimension, any V.
- $d \geq 1$ dimensions, $V(u) = au^2$.



WHILE FUNAKI-SPOHN (1997) SHOWED THAT THE SURFACE TENSION

$$\sigma_V(u_1, u_2) := -\lim_{N\uparrow\infty} \frac{1}{N^2} \log Z_N(u_1, u_2)$$

EXISTS WHEN $0 < c_1 \leq V''(u) \leq c_2 < \infty$, WE ONLY HAVE FORMULAS IN TWO SPECIAL CASES:

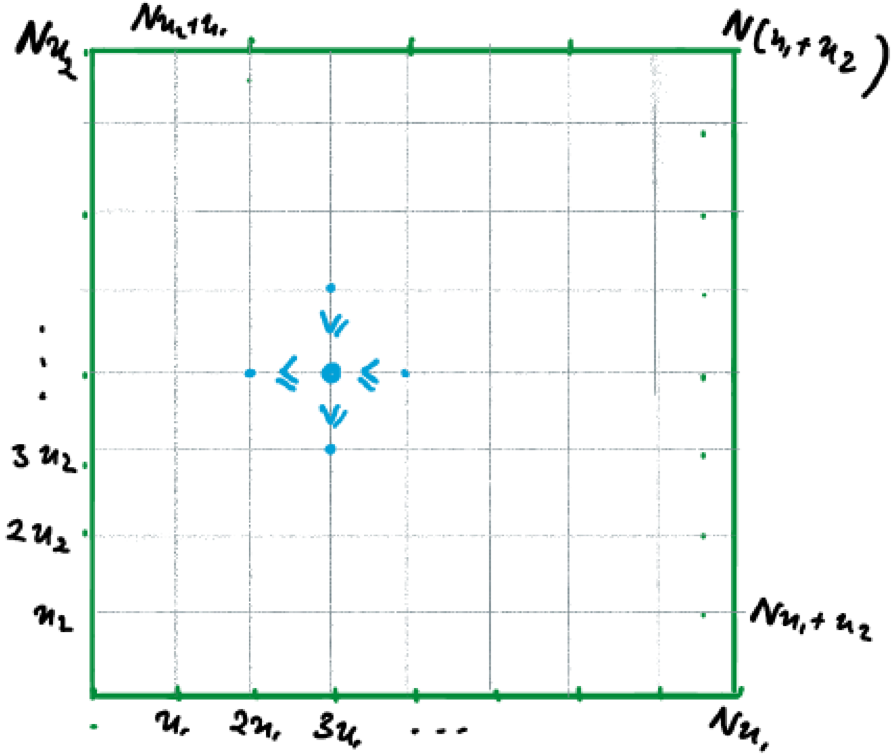
- One dimension, any V.
- $d \geq 1$ dimensions, $V(u) = au^2$.

What about $V(u) = \begin{cases} \infty & u < 0 \\ 0 & u \geq 0 \end{cases}$ in $d=2$?

WE WANT TO CALCULATE SURFACE TENSION OF BEAD MODEL.

EQUIVALENTLY, CAN WE CALCULATE

$$\sigma(u_1,u_2) = -\lim_{N\uparrow\infty} \frac{1}{N^2} \log Z_N(u_1,u_2),$$



WHERE

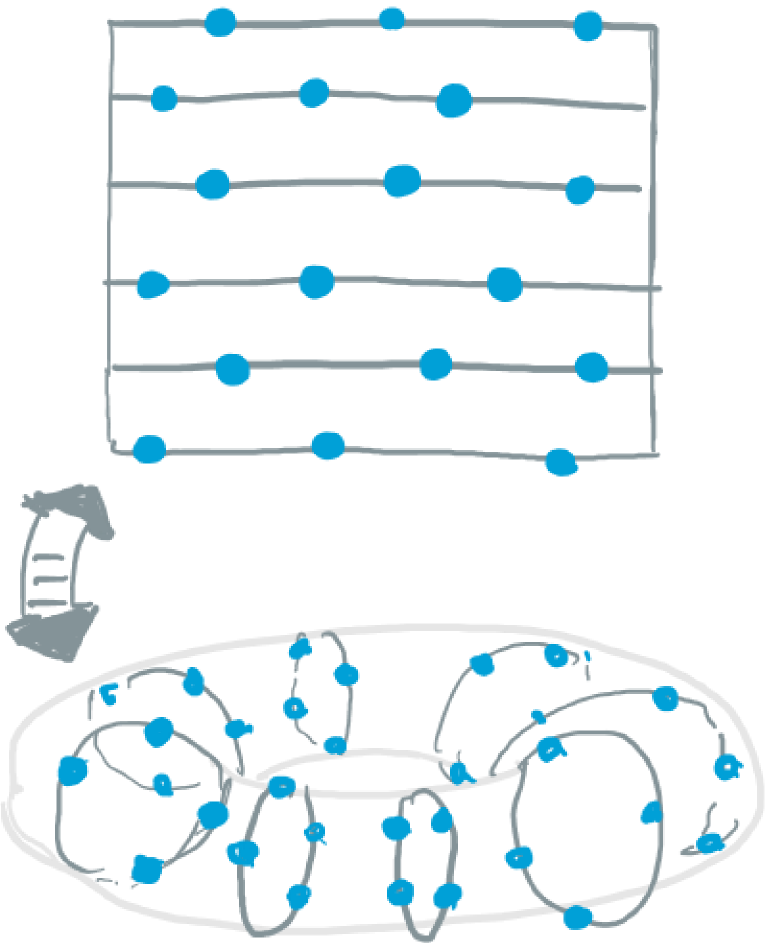
$$Z_N(u_1,u_2) := \int_{\mathbb{R}^{(N-1)^2}} \prod_{1\le i,j\le N-1} d\phi_{ij} \; \mathbb{1}\left[\begin{matrix}\phi_{i-1,j} \\ \phi_{i,j-1}\end{matrix} \leqslant \phi_{ij} \leqslant \begin{matrix}\phi_{i+1,j} \\ \phi_{i,j+1}\end{matrix} \right]$$

WITH BOUNDARY CONDITIONS

$$\phi_{ij} = iu_1 + ju_2$$

WHENEVER $i \in \{0,N\}$ OR $j \in \{0,N\}$.

WE CAN USE A RESULT FROM J. 2022:



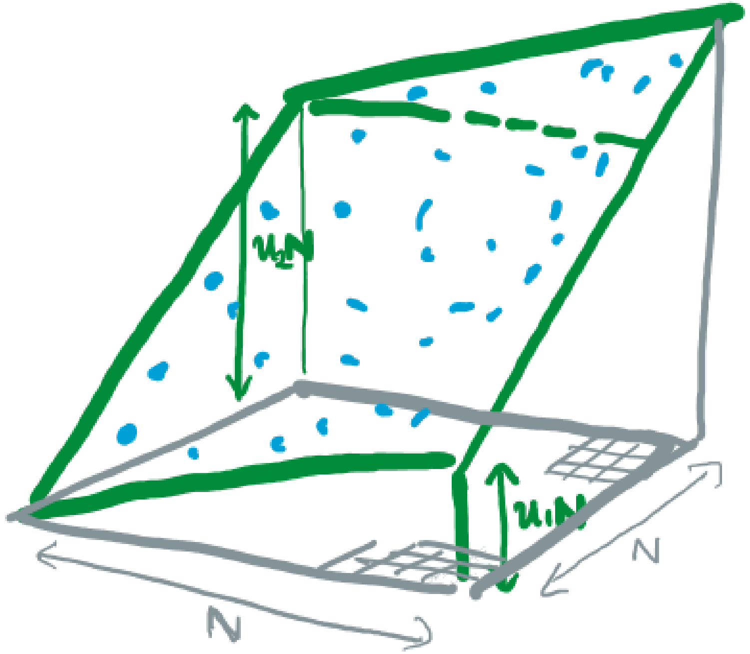
EXPLICIT FORMULA FOR VOLUME OF BEAD CONFIGURATIONS ON A TORUS. WITH

n UNIT LENGTH STRINGS

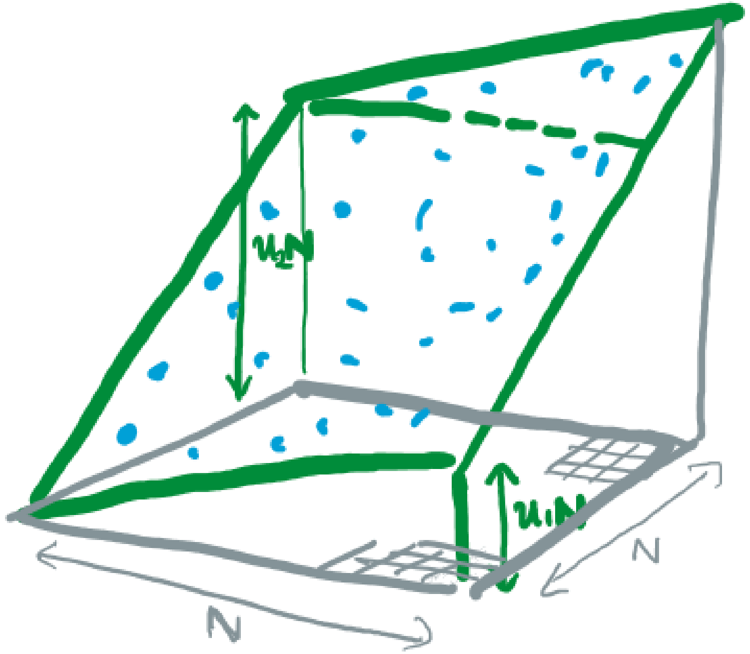
k BEADS PER STRING.

SEE ALSO:

- SUN (2018)
- J. + O'CONNELL (2020)
- GORDENKO (2021).



$$\sigma_{V_{GT}}(u_1, u_2) = -\lim_{N\uparrow\infty} \frac{1}{N^2} \log \left[\begin{array}{l} \text{volume of increasing functions} \\ \text{on square } \square_N = \{0,1,\dots,N\}^2 \\ \text{WITH LINEAR BOUNDARY SLOPES } u_1, u_2 \end{array} \right]$$



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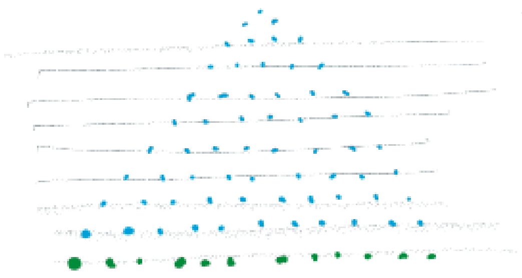
THEOREM (J.+ PROCHNO 2024)

THE SURFACE TENSION ASSOCIATED WITH $V(u) = \infty 1\{u<0\}$ EXISTS AND IS GIVEN BY

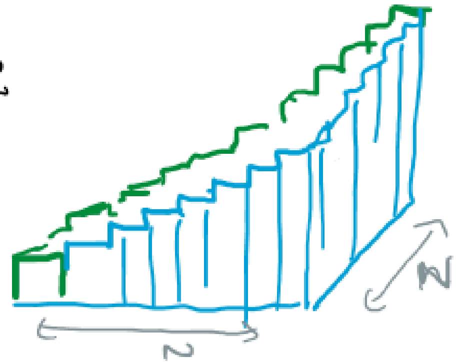
$$\sigma_{V_{GT}}(u_1,u_2) = \begin{cases} -\log(u_1+u_2) - \log\sin\frac{\pi u_1}{u_1+u_2} - 1 + \log\pi & u_1,u_2>0 \\ +\infty & \text{otherwise} \end{cases}$$



RANDOM GELFAND TSETLIN PATTEN WITH BOTTOM ROW

$\lambda_1 \leq \cdots \leq \lambda_N$ $\quad \frac{1}{N}\sum_{i=1}^{N} \delta_{\lambda_i} \approx \mu$

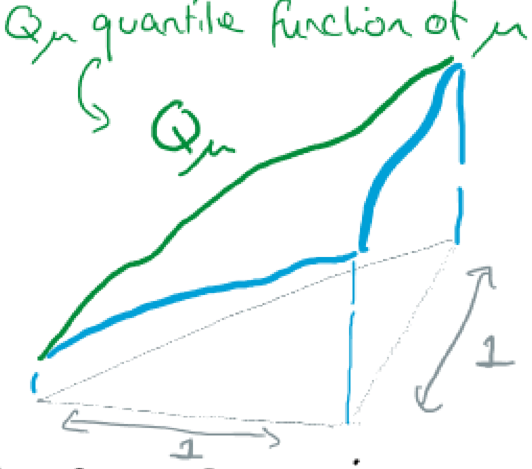
TURN INTO FUNCTION



RANDOM INCREASING SURFACE

$\phi : \Delta_N \to \mathbb{R}$.

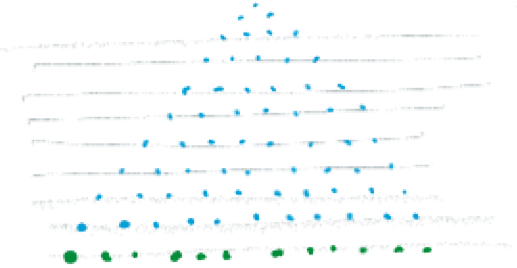
RESCALE $1/N$



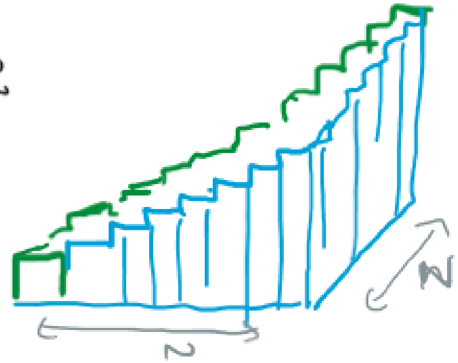
RANDOM FUNCTION

$\tilde{\phi} : \Delta \to \mathbb{R}$.

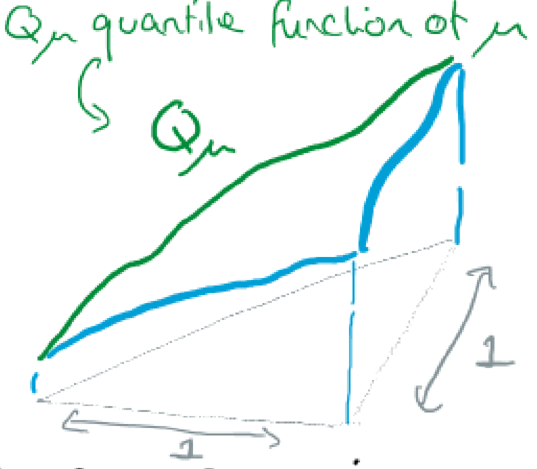
$\Delta = \{0 \leq t \leq s \leq 1\}$.



TURN INTO FUNCTION $\longrightarrow$



RESCALE $1/N$ $\longrightarrow$



RANDOM GELFAND TSETLIN PATTEN WITH BOTTOM ROW $\lambda_1 \leq \cdots \leq \lambda_N$ $\quad \frac{1}{N}\sum_{i=1}^{N} \delta_{\lambda_i} \approx \mu$

RANDOM INCREASING SURFACE $\phi : \Delta_N \to \mathbb{R}$.

RANDOM FUNCTION $\tilde{\phi} : \Delta \to \mathbb{R}$.

$\Delta = \{ 0 \leq t \leq s \leq 1 \}$.

THEOREM (J. + PROCHNO 2024)

- THE RESCALED RANDOM FUNCTION ASSOCIATED WITH A GIANT RANDOM GELFAND TSETLIN PATTERN WITH BOTTOM ROW $\approx \mu$ CONVERGES a.s. in L^2 TO THE MINIMISER OF

$$\mathcal{E}[\psi] = \int_\Delta \sigma(\nabla \psi)\, ds\, dt, \quad \sigma(u_1, u_2) = -\log(u_1 + u_2) - \log \sin \frac{\pi u_1}{u_1 + u_2} - 1 + \log \pi$$

SUBJECT TO BOUNDARY CONDITION $\psi = Q_\mu$ ON DIAGONAL.

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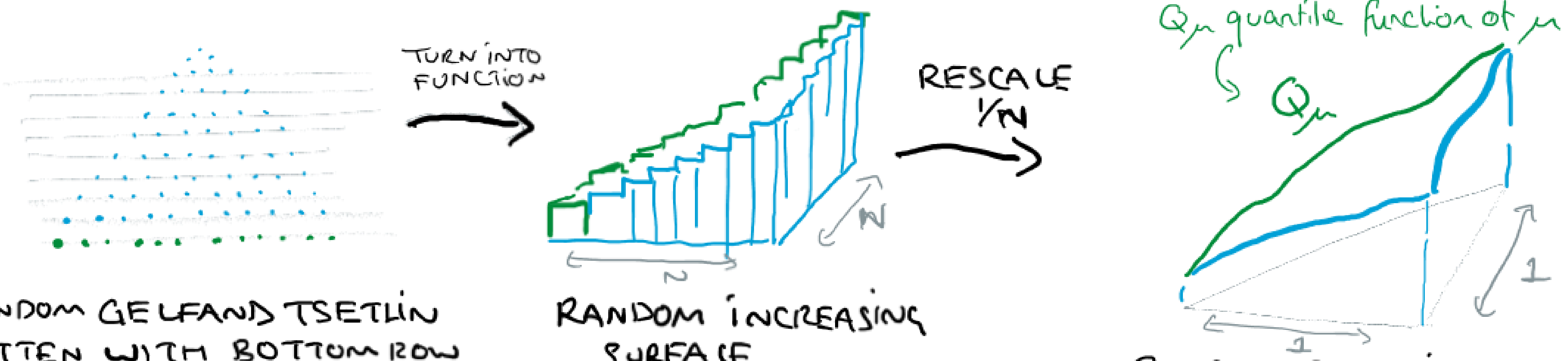

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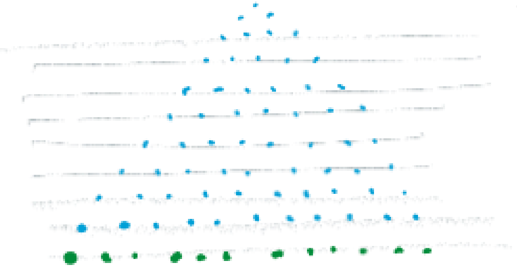
- THE MINIMISER $\underline{\underline{\text{IS}}}$ FREE COMPRESSION.



RANDOM GELFAND TSETLIN PATTEN WITH BOTTOM ROW

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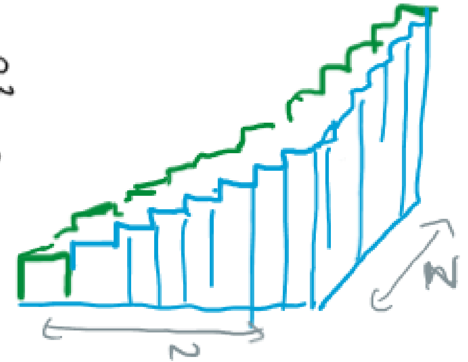
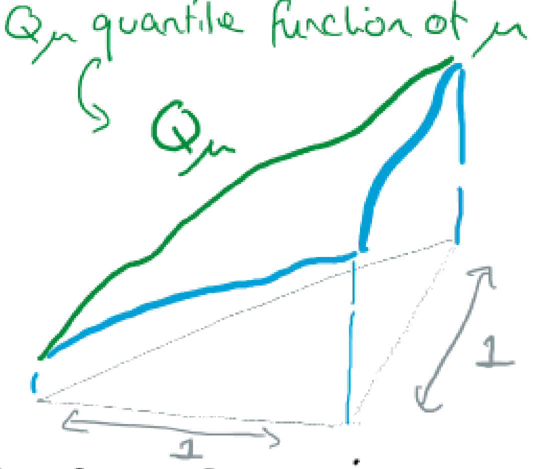
TURN INTO FUNCTION $\longrightarrow$



RANDOM INCREASING SURFACE

$\phi : \Delta_N \to \mathbb{R}.$

RESCALE $1/N$ $\longrightarrow$



RANDOM FUNCTION

$\tilde{\phi} : \Delta \to \mathbb{R}.$

$\Delta = \{0 \leq t \leq s \leq 1\}.$

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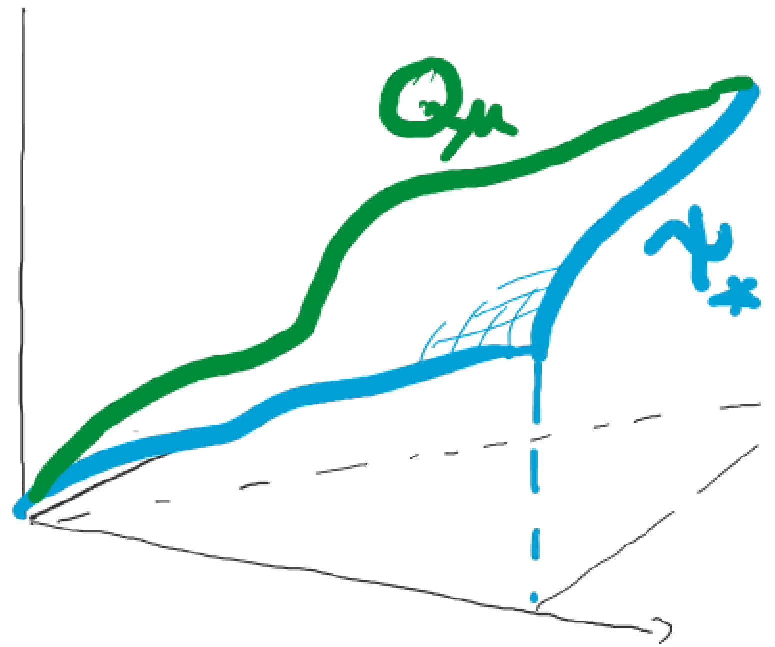
SUBJECT TO BOUNDARY CONDITION $\psi = Q_\mu$ ON DIAGONAL.

- THE MINIMISER <u>IS</u> FREE COMPRESSION, EXPLAINING SHLYAKHTENKO + TAO'S EULER LAGRANGE EQUATION.

USUALLY OUR RANDOM SURFACE CONCENTRATES AROUND THE MINIMISER $\psi_*: \Delta \to \mathbb{R}$ OF



$$E[\psi] := \int_\Delta \sigma(\nabla \psi)\, ds\, dt .$$

SUBJECT TO BOUNDARY CONDITIONS.

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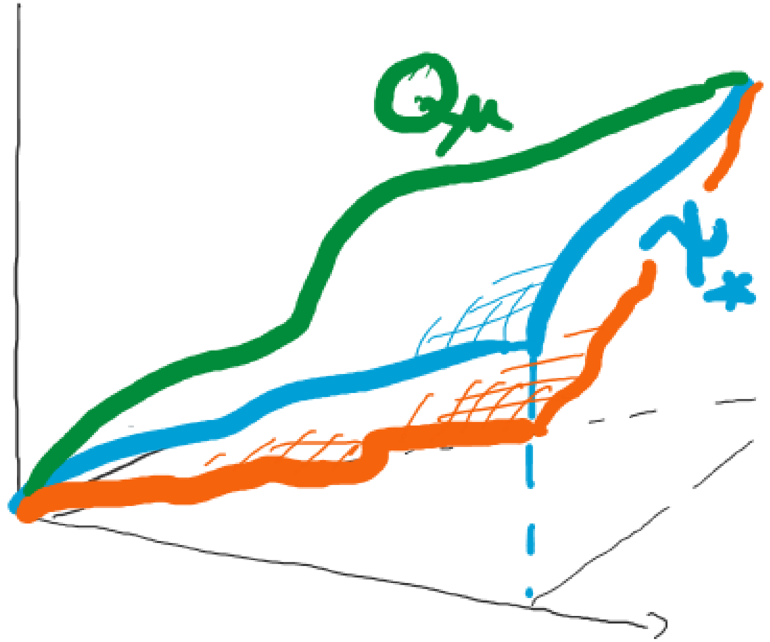
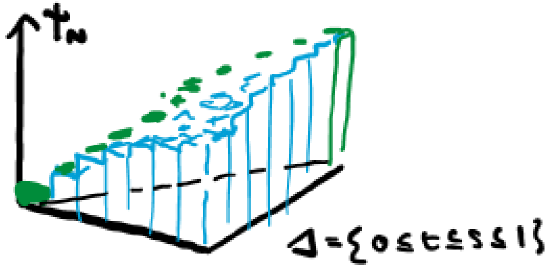


$$E[\psi] := \int_\Delta \sigma(\nabla \psi)\, ds\, dt .$$

SUBJECT TO BOUNDARY CONDITIONS.

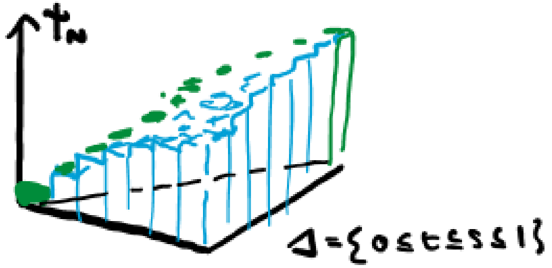
BUT WHAT ABOUT RARE EVENTS?



$$\mathcal{L}^1_{inc}(\Delta) = \left\{ \begin{array}{l} \psi : \Delta \to \mathbb{R} \text{ in } \mathcal{L}^1, \\ \text{increasing in both} \\ \text{coordinates} \end{array} \right\}.$$

WE CAN ASSOCIATE A GELFAND-TSETLIN PATTERN WITH AN ELEMENT OF THE METRIC SPACE $\mathcal{L}^1_{inc}(\Delta)$ WITH $\mathcal{L}^1$ METRIC:

$$d(f,g) := \int_\Delta |f(s,t) - g(s,t)| \, ds \, dt$$

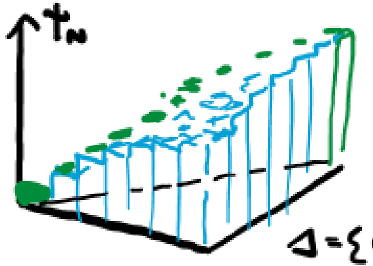


$$\mathcal{L}^1_{inc}(\Delta) = \left\{ \psi : \Delta \to \mathbb{R} \text{ in } L^1, \text{ increasing in both coordinates} \right\}.$$

THEOREM (J. + PROCHNO 2024)

- THE RESCALED RANDOM FUNCTION ψ_N ASSOCIATED WITH A GIANT RANDOM GELFAND TSETLIN PATTERN WITH BOTTOM ROW $\approx \mu$ SATISFIES A **LARGE DEVIATION PRINCIPLE** IN $\mathcal{L}^1_{inc}(\Delta)$ WITH SPEED N^2 AND RATE FUNCTION

$$I_\mu[\psi] = \int_\Delta \sigma(\nabla\psi)\,ds\,dt + \chi_\mu. \quad \text{SUBJECT TO } \psi = Q_\mu.$$



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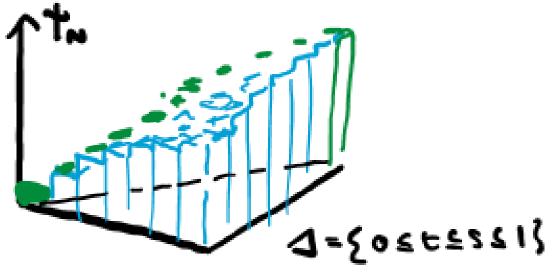
INFORMALLY,
$$\mathbb{P}(\psi_N \in \Gamma) \approx e^{-N^2 \inf_{\psi \in \Gamma} I_\mu[\psi]}.$$



$$\mathcal{L}^1_{inc}(\Delta) = \left\{ \Psi : \Delta \to \mathbb{R} \text{ in } L^1, \text{ increasing in both coordinates} \right\}.$$

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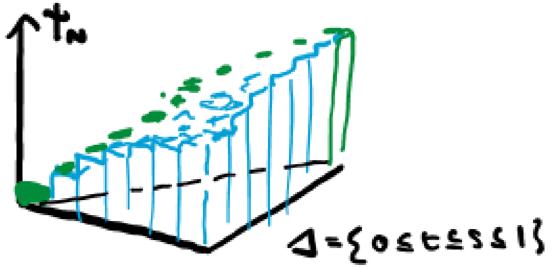
$$I_\mu[\Psi] = \int_\Delta \sigma(\nabla \Psi)\, ds\, dt + \mathcal{X}_\mu. \quad \text{SUBJECT TO } \Psi = Q_\mu.$$

INFORMALLY, $\quad \mathbb{P}(\Psi_N \in \Gamma) \approx e^{-N^2 \inf_{\Psi \in \Gamma} I_\mu[\Psi]}.$

FORMALLY

$$-\inf_{\Psi \in \Gamma^\circ} I_\mu[\Psi] \le \liminf_{N \to \infty} \frac{1}{N^2} \log \mathbb{P}(\Psi_N \in \Gamma) \le \limsup_{N \to \infty} \frac{1}{N^2} \log \mathbb{P}(\Psi_N \in \Gamma) \le -\inf_{\Psi \in \overline{\Gamma}} I_\mu[\Psi]$$



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FORMALLY

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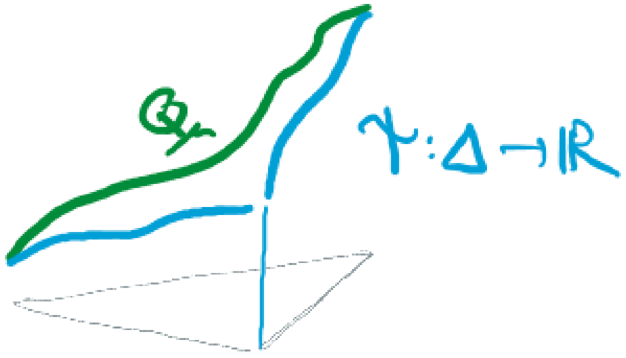
RECALL $I_\mu[\psi] = \int_\Delta \sigma(\nabla\psi)\,ds\,dt + \chi_\mu$ IS MINIMISED OVER ALL SURFACES BY $\psi_* =$ FREE COMPRESSION, AND $I_\mu[\psi_*] = 0$.



$Q_\mu =$ quantile function of μ.

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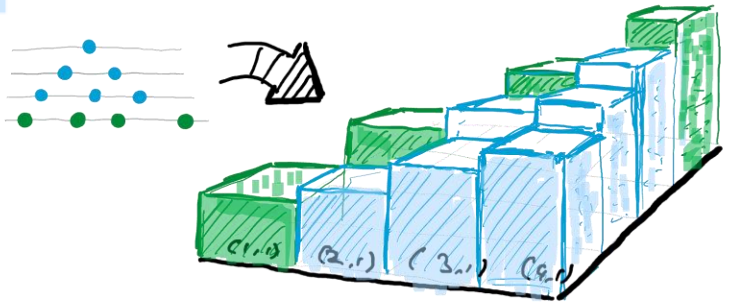
$Q_\mu =$ quantile function of μ.

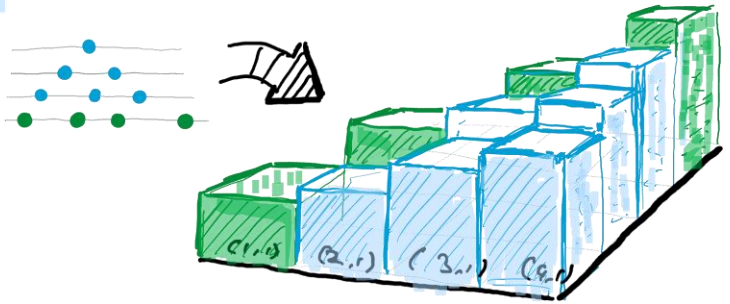
THEOREM (J. + PROCHNO 2024)

$$\chi_\mu = \int_\Delta \sigma(\nabla\psi_*)\,ds\,dt = \frac{1}{2}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \log|y-x|\,\mu(dy)\,\mu(dx) + 3/4 .$$

(WHICH IS BETTER KNOWN AS "VOICULESCU'S FREE ENTROPY")

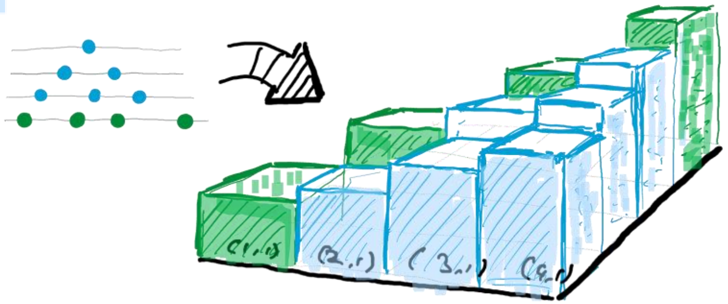


A GELFAND-TSETLIN PATTERN IS THE SAME AS AN INCREASING FUNCTION $\phi : \Delta_n \to \mathbb{R}$



A GELFAND-TSETLIN PATTERN IS THE SAME AS AN INCREASING FUNCTION $\phi : \Delta_n \to \mathbb{R}$

A LARGE G-T PATTERN WITH BOTTOM ROW APPROXIMATING μ GIVES RISE TO AN INCREASING FUNCTION $\psi : \Delta \to \mathbb{R}$ WITH DIAGONAL $\approx Q_\mu$.



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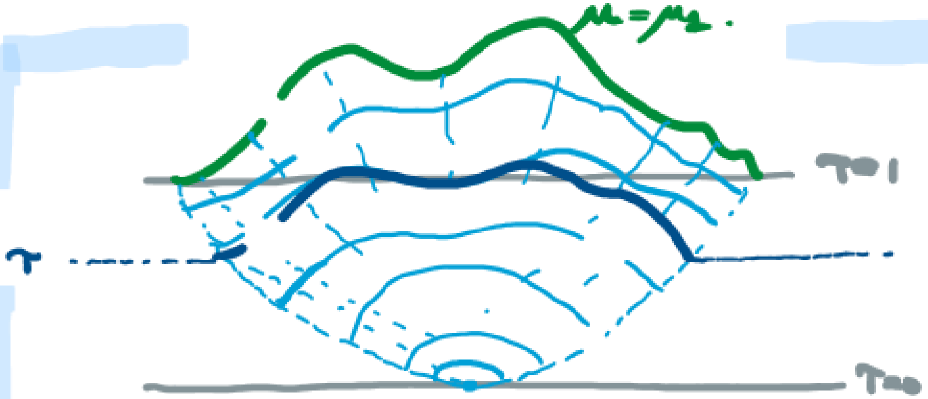
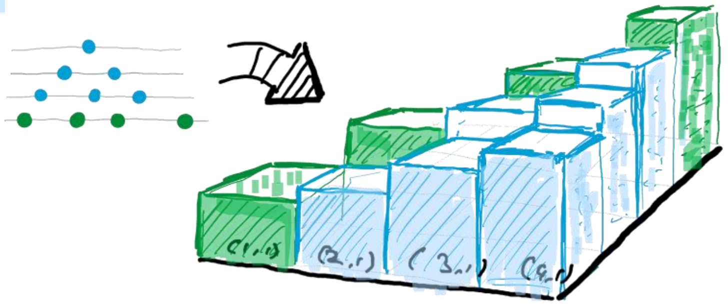


CALCULATIONS IN FREE PROBABILITY GIVE A DESCRIPTION OF SHAPE OF LARGE RANDOM G-T PATTERNS



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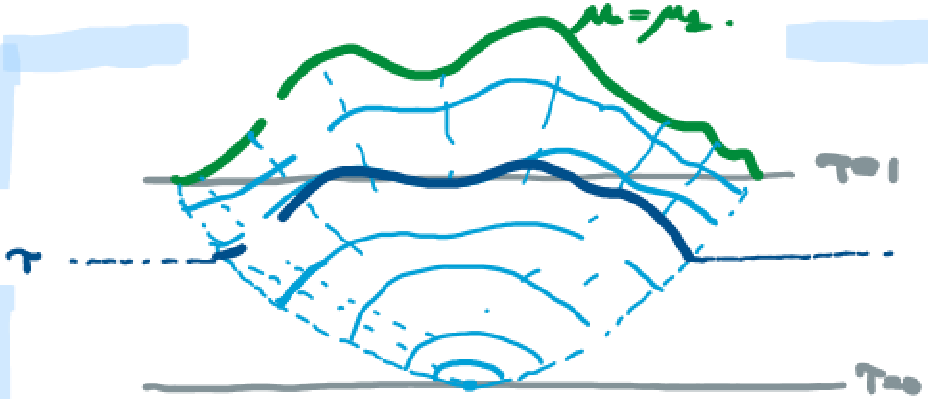


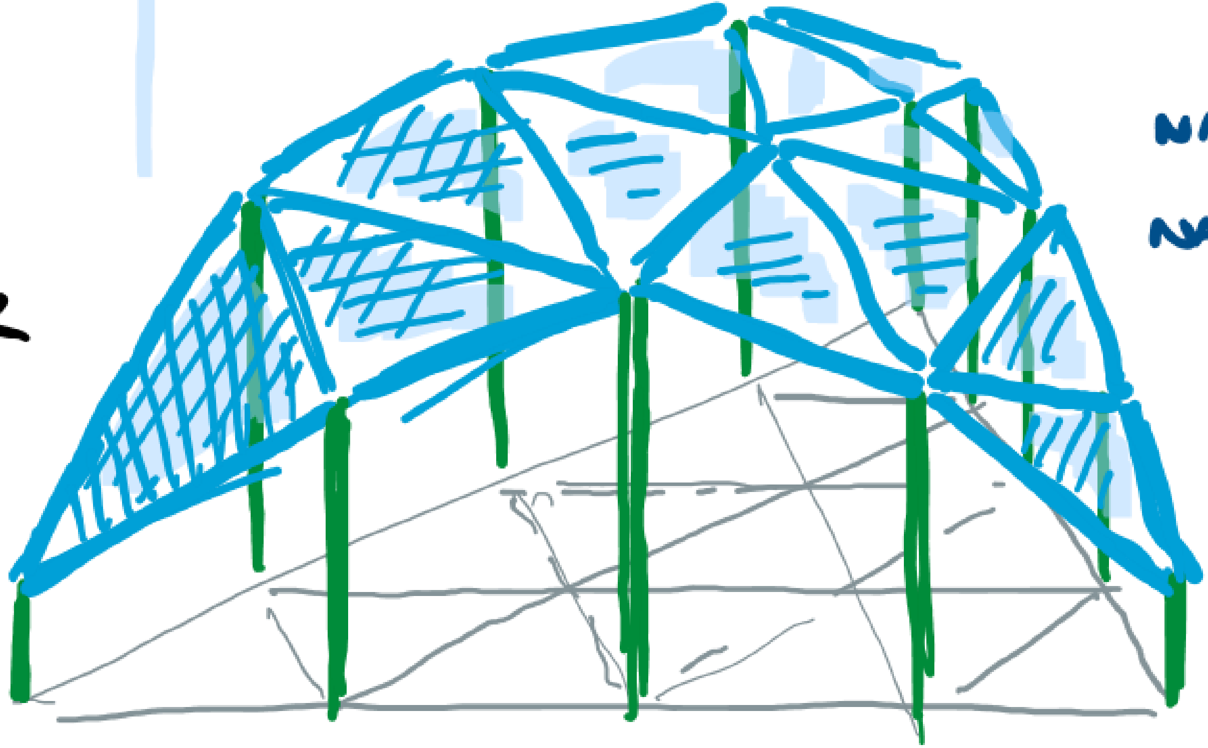
CALCULATIONS IN FREE PROBABILITY GIVE A DESCRIPTION OF SHAPE OF LARGE RANDOM G-T PATTERNS

BUT WE GIVE A NEW ROUTE TO THIS FACT (+ LDP) USING STATISTICAL PHYSICS OF RANDOM SURFACES, PROVING SHLYAKHTENKO + TAO'S CONJECTURE.

WHAT ABOUT THE HIVE?

hive (noun)

A CONCAVE FUNCTION DEFINED ON A TRIANGULAR LATTICE



NARAYANAN – SHEFFIELD 2023

NARAYANAN – SHEFFIELD – TAO 2024.

THANK YOU FOR LISTENING !